

Variance Estimation from SAS Poulpe to R Gustave

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1 Introduction

This vignette documents the migration of variance estimation workflows from the SAS Poulpe macro to R, using the `gustave` package developed by INSEE. This vignette is intended for survey methodologists migrating from the SAS Poulpe macro to R, or for methodologists who need to implement analytical variance estimation for complex surveys.

The core outcome is a set of **variance estimation wrappers**: self-contained R functions that encapsulate the full sampling methodology (stratification, non-response correction, calibration, multi-stage sampling) and can be distributed to thematic units. Once a wrapper has been prepared by the methodologist, analysts can compute precision measures on their own datasets with a single function call — without needing to understand the underlying formulas. Wrappers can also be called from Python through `rpy2`, allowing seamless integration into existing analytical pipelines.

We have validated the R implementation against **Poulpe** reference values across five progressively complex cases, from simple stratified sampling to two-stage designs with individual-level estimation. The results match, confirming that **gustave** fully replaces **Poulpe** for the survey designs used at Statbel.

Beyond replication, this migration brings three practical benefits. First, the variance functions are transparent and version-controlled, making them auditable and reproducible. Second, custom **qvar**-like functions standardise wrapper creation and reduce the risk of methodological errors. Third, diagnostic tools and batch estimation functions streamline quality reporting and production workflows.

The remainder of this vignette is structured as follows. Sections 2 through 6 present the five cases with their mathematical formulations, R implementations, and **Poulpe** comparisons. Section 7 covers the practical aspects of building production-ready wrappers. Section 8 concludes with limitations and outlook.

1.1 Why variance estimation?

Survey estimates are based on a sample, not the full population. As such, they are subject to **sampling variability**: a different sample would yield different results. Variance estimation quantifies this uncertainty, and is the foundation for computing confidence intervals, coefficients of variation, and significance tests.

In practice, our surveys involve several layers of complexity beyond simple random sampling: **stratification**, **non-response correction** (through reweighting), and **calibration** on known population margins. Each of these steps affects the variance, and ignoring them leads to biased precision measures.

1.2 The **gustave** package

gustave (a User-oriented Statistical Toolkit for Analytical Variance Estimation) is an R package developed by INSEE. It provides:

- Standard analytical variance estimators (Sen-Yates-Grundy, Deville-Tillé).
- A mechanism to produce **variance estimation wrappers**: user-friendly functions that encapsulate the full estimation methodology.
- Built-in handling of linearization (for means, ratios, etc.) and domain estimation.

gustave is the R successor to the **Poulpe** SAS macro, also developed by INSEE and historically used at Statbel. Throughout this vignette, we provide comparisons between the two tools to facilitate the transition.

1.3 Structure of this vignette

We build up the variance estimation progressively, from the simplest to the most complex case:

1. **Stratified simple random sampling** (no non-response, no calibration)
2. **Adding non-response** correction through reweighting
3. **Adding calibration** on population margins
4. **Two-stage sampling** with area then dwelling
5. **Two-stage sampling** with area then dwelling, with individual answers

At each step, we introduce the relevant **gustave** functions and concepts:

- **qvar()**: a quick, ready-to-use function for common survey designs.
- **define_variance_wrapper()**: the core function for defining custom variance estimation wrappers, necessary for more complex designs.
- The underlying variance functions: **var_srs()**, **var_pois()**, **res_cal()**, **add_zero()**.

We also introduce new functions developed in-house to help us to operationalise the variance estimation :

- `inspect_wrapper()`: a function to run diagnostics on a variance wrapper, by extracting the reference population, the technical data inventory, and the variance estimation pipeline.
- `call_gustave()`: a function to call a **gustave** variance wrapper in a different way.
- `batch_gustave()`: a function to estimate variance across indicators, groups, and domains (like the legacy `boucle_poulpe`).

1.4 Environment

```
library(tidyverse)
library(gt)
library(gustave)

lfs_samp_area <- gustave::lfs_samp_area
lfs_samp_dwel <- gustave::lfs_samp_dwel
lfs_samp_ind <- gustave::lfs_samp_ind
```

1.5 Data

The vignette uses a simulated Labour Force Survey shipped with the **gustave** package: a two-stage design — areas are drawn as primary sampling units, dwellings within areas, and all individuals of a sampled dwelling are surveyed. The product of the two inclusion probabilities is constant across dwellings ($\text{pik} = \text{pik_area} * \text{pik_dwel} \approx 0.0083$), so that the one-stage weight $1/\text{pik}$ matches the individual-level `sampling_weight`. There is no non-response into the data, we will create it later on purpose.

Key variables in the area sample (`lfs_samp_area`, 4 areas):

- `id_area`: area identifier (primary sampling unit)
- `income`: area-level income aggregate (illustrative; the estimations below use the dwelling- and individual-level incomes)
- `pik_area`: probability of selection of the area (between 0.0499 and 0.0503)

Key variables in the dwelling sample (`lfs_samp_dwel`, 80 dwellings in 4 areas, 20 dwellings per area on average):

- `id_dwel`: dwelling identifier
- `id_area`: area identifier of the dwelling
- `income`: dwelling income (between 2.5 and 46.4)
- `pik_area`: probability of selection of the area
- `pik_dwel`: probability of selection of the dwelling within its area (between 0.165 and 0.167)
- `pik`: overall probability of selection of the dwelling ($\text{pik_area} * \text{pik_dwel}$, constant ≈ 0.0083)

Key variables in the individual sample (`lfs_samp_ind`, 116 individuals):

- `id_ind`: individual identifier
- `id_dwel`: dwelling identifier of the individual all individuals of a sampled dwelling are surveyed (exhaustive stage)
- `income`: individual income
- `unemp`: unemployment indicator (7.8 % of individuals)

- **sampling_weight**: overall sampling weight of the individual ($1/\text{pik}$ of its dwelling, ≈ 120.24)

The vignette preamble derives the following variables. For the sampling design: $w_{\text{dwel}} = 1/\text{pik}_{\text{dwel}}$ (conditional dwelling weight, ≈ 6), $w_{\text{area}} = \text{mean}(1/\text{pik}_{\text{area}})$ (conditional area weight, equalized across areas as the SRS convention requires, ≈ 19.9), and $w_{\text{final}} = w_{\text{dwel}} * w_{\text{area}}$ (two-stage design weight, ≈ 120).

```
lfs_samp_area$w_area <- 1 / lfs_samp_area$pik_area
lfs_samp_area$w_area <- base::mean(lfs_samp_area$w_area)
lfs_samp_dwel$w_dwel <- 1 / lfs_samp_dwel$pik_dwel
lfs_samp_dwel$w_final <- lfs_samp_area$w_area * lfs_samp_dwel$w_dwel
lfs_samp_dwel <- lfs_samp_dwel %>% select(-pik_area) %>%
  left_join(lfs_samp_area %>% select(id_area, w_area), by = "id_area")
```

For the first three cases, we will consider only the dwelling level to estimate the variance. Here we consider a stratified random sampling of dwellings into population by area, and we only use the dwelling answers, here the income for the example.

gustave needs a Boolean variable to indicate if the unit must be present for dissemination. Here all unit are used.

```
lfs_samp_dwel$diss <- TRUE
```

2 Case 1: Stratified Simple Random Sampling

2.1 Context

In this first case, we assume **full response** and **no calibration**. We estimate the variance of the total income under a stratified simple random sampling design of dwellings, where the strata are defined by **id_area**.

This is the simplest case and allows us to understand the fundamental mechanics of **gustave**.

2.2 Variance decomposition

Before using **gustave**, let us verify the variance formula manually. Under stratified SRS without replacement, the variance of the estimated total $\hat{Y}$ is:

$$V(\hat{Y}) = \sum_{h=1}^H N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}$$

where N_h and n_h are the population and sample sizes in stratum h , and s_h^2 is the sample variance within stratum h .

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i \in h} (y_i - \bar{y}_h)^2$$

The variance estimate can be calculated manually in R:

```

lfs_samp_dwel %>%
  replace_na(list(income = 0)) %>%
  group_by(id_area) %>%
  mutate(
    N_h = sum(w_final),
    n_h = n(),
    y_bar = base::mean(income),
    dev2 = (income - y_bar)^2
  ) %>%
  summarise(
    N_h = base::mean(N_h),
    n_h = base::mean(n_h),
    s2 = sum(dev2) / (base::mean(n_h) - 1),
    .groups = "drop"
  ) %>%
  mutate(
    var_h = N_h^2 * (1 - n_h / N_h) * s2 / n_h
  ) %>%
  summarise(
    var_total = sum(var_h),
    se_total = sqrt(sum(var_h))
  )
#> # A tibble: 1 x 2
#>   var_total se_total
#>   <dbl>     <dbl>
#> 1 120629899.  10983.

```

In *gustave*, this variance component is obtained with the function `var_srs` : `var_srs(y,  $\pi_i$ , strata)`.

2.3 Using `qvar()`

`qvar()` is the quick entry point into *gustave*. It handles the most common survey designs in a single function call.

2.3.1 Direct estimation

When called with a statistic (here `total(income)`), `qvar()` performs the variance estimation directly:

```

qvar(
  data = lfs_samp_dwel,
  id = "id_dwel",
  dissemination_dummy = "diss",
  dissemination_weight = "w_final",
  sampling_weight = "w_final",
  strata = "id_area",
  total(income)
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - stratified simple random sampling
#> Note: The strata variable (id_area) is of type character. It is automatically coerced
#>   to factor.

```

```
#>           call      n      est variance      std      cv lower  upper
#> 1 total(y = income) 80 198343.6 120629899 10983.16 5.537443 176817 219870.2
```

The key arguments are:

- `id`: the unit identifier (must be character).
- `dissemination_dummy`: a logical or 0/1 variable indicating which units are disseminated (here, all units).
- `dissemination_weight`: the final weight used for point estimation.
- `sampling_weight`: the initial design weight.
- `strata`: the stratification variable.

`qvar()` displays notes about the variance estimation process, recalling the survey design and the variable transformations it required. The output shows the estimated total (`est`), the estimated variance (`var`), the standard error (`std`), the coefficient of variation (`cv`), and the lower and upper bounds of a 95% confidence interval (`lower` and `upper`).

2.3.2 Defining a variance wrapper

When you need to estimate the variance for several variables, it is more efficient to define a **variance wrapper** once and re-use it. This is done by adding `define = TRUE` to `qvar()`:

```
precision <- qvar(
  data = lfs_samp_dwel,
  id = "id_dwel",
  dissemination_dummy = "diss",
  dissemination_weight = "w_final",
  sampling_weight = "w_final",
  strata = "id_area",
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - stratified simple random sampling
#> Note: The strata variable (id_area) is of type character. It is automatically coerced
#>   to factor.
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

The resulting `precision` object is a function. It contains all the methodological information and technical data needed for variance estimation. The end user simply calls it with the dataset and the statistic(s) of interest:

```
# Variance of the total income
precision(lfs_samp_dwel, total(income))
#>           call      n      est variance      std      cv lower  upper
#> 1 total(y = income) 80 198343.6 120629899 10983.16 5.537443 176817 219870.2

# Several statistics at once, with labels
precision(lfs_samp_dwel,
  "Total income" = total(income),
```

```

"Mean income" = mean(income)
)
#>           label           call  n           est      variance           std
#> 1 Total income total(y = income) 80 198343.61297 1.206299e+08 10983.164353
#> 2 Mean income mean(y = income) 80    20.61972 1.303719e+00    1.141805
#>           cv           lower           upper
#> 1 5.537443 176817.00640 219870.21954
#> 2 5.537443    18.38182    22.85761

# Statistics with filter
precision(lfs_samp_dwel, mean(income), where = id_area == "A025")
#>           call  n           est variance           std
#> 1 mean(y = income, where = id_area == "A025") 20 25.18193 5.133222 2.265662
#>           cv           lower           upper
#> 1 8.997172 20.74132 29.62255

# Statistics by groups
precision(lfs_samp_dwel, mean(income), by = id_area)
#>           call  by  n           est variance           std           cv
#> 1 mean(y = income, by = id_area) A005 20 15.60537 5.942796 2.437785 15.621452
#> 2 mean(y = income, by = id_area) A007 20 17.26456 4.465462 2.113164 12.239891
#> 3 mean(y = income, by = id_area) A016 20 24.45873 5.319474 2.306399 9.429756
#> 4 mean(y = income, by = id_area) A025 20 25.18193 5.133222 2.265662 8.997172
#>           lower           upper
#> 1 10.82740 20.38334
#> 2 13.12284 21.40629
#> 3 19.93827 28.97919
#> 4 20.74132 29.62255

```

This is the core philosophy of **gustave**: the **methodologist** defines the wrapper (once), and the **analyst** uses it (repeatedly) without needing to understand the underlying variance estimation methodology. The dataset used by the **analyst** needs only the id variable of dwelling, everything else comes from the wrapper.

The wrapper can be saved as an `.rds` file and distributed to thematic units. They only need to load the **gustave** package and the wrapper object — from there, they can compute variance estimates on their own dataset. The only constraint is that the identifier variable in their dataset must match the one used when defining the wrapper.

```

saveRDS(precision, file = "precision_dwel.rds")

rm(precision)

library(gustave)
precision <- readRDS(file="precision_dwel.rds")

# lfs_samp_dwel could be another dataset with other indicators
precision(lfs_samp_dwel, total(income))
#>           call  n           est variance           std           cv lower           upper
#> 1 total(y = income) 80 198343.6 120629899 10983.16 5.537443 176817 219870.2

```

2.4 Using `define_variance_wrapper()`

`qvar()` covers the most common cases, but for complex survey designs (multi-stage sampling, balanced sampling, custom non-response corrections), we need the more flexible `define_variance_wrapper()`.

To use it, we must first write a **variance function**: an R function that takes a matrix `y` of the variable of interest and returns the estimated variance.

2.4.1 Variance function

The variance function receives `y` as a named matrix (with row names matching the unit identifiers) and accesses the technical data through its enclosing environment.

For our stratified SRS case, we only need `var_srs()`:

```
var_sample <- function(y, lfs_samp_dwel) {  
  variance <- list()  
  variance[["srs"]] <- var_srs(  
    y,  
    pik = 1 / lfs_samp_dwel$w_final,  
    strata = lfs_samp_dwel$id_area  
  )  
  Reduce(`+`, variance)  
}
```

A few things to note:

- `lfs_samp_dwel` is not, here, the dwelling dataset. It is only a parameter for the function. We use the same name for convenience.
- `var_srs()` implements the variance estimator for stratified SRS.
- `pik` is the vector of first-order inclusion probabilities.
- We store variance components in a named list. This pattern will become useful later when we add more components (non-response, calibration).

2.4.2 Defining the wrapper

To create the wrapper, we need to give technical data (here the dataset with all information about the sampling and the dissemination weight).

```
technical_data <- list()  
technical_data$lfs_samp_dwel <- lfs_samp_dwel  
  
precision2 <- define_variance_wrapper(  
  variance_function = var_sample,  
  technical_data = technical_data,  
  reference_id = technical_data$lfs_samp_dwel$id_dwel,  
  reference_weight = technical_data$lfs_samp_dwel$w_final,  
  default_id = "id_dwel"  
)
```

The arguments are:

- `variance_function`: our custom variance function.
- `technical_data`: a list containing all auxiliary data needed by the variance function (sampling weights, strata, response probabilities, etc.). This data is embedded in the wrapper.
- `reference_id`: the vector of unit identifiers in the reference population (here, all sampled units).
- `reference_weight`: the dissemination weights for the reference population.
- `default_id`: the name of the identifier column in the dataset passed at estimation time.

2.4.3 Using the wrapper

Using the wrapper is very easy, we only need to give it a dataset (with a list of units indexed by the same id as the technical data). The result should match `qvar()` exactly.

```
precision2(lfs_samp_dwel, total(income))
#>           call n      est variance      std      cv lower  upper
#> 1 total(y = income) 80 198343.6 120629899 10983.16 5.537443 176817 219870.2

# Several statistics at once, with labels
precision2(lfs_samp_dwel,
  "Total income" = total(income),
  "Mean income"  = mean(income)
)
#>           label           call n      est variance      std
#> 1 Total income total(y = income) 80 198343.61297 1.206299e+08 10983.164353
#> 2 Mean income mean(y = income) 80      20.61972 1.303719e+00      1.141805
#>           cv      lower      upper
#> 1 5.537443 176817.00640 219870.21954
#> 2 5.537443      18.38182      22.85761

# Statistics with filter
precision2(lfs_samp_dwel, mean(income), where = id_area == "A025")
#>           call n      est variance      std
#> 1 mean(y = income, where = id_area == "A025") 20 25.18193 5.133222 2.265662
#>           cv      lower      upper
#> 1 8.997172 20.74132 29.62255

# Statistics by groups
precision2(lfs_samp_dwel, mean(income), by = id_area)
#>           call by n      est variance      std      cv
#> 1 mean(y = income, by = id_area) A005 20 15.60537 5.942796 2.437785 15.621452
#> 2 mean(y = income, by = id_area) A007 20 17.26456 4.465462 2.113164 12.239891
#> 3 mean(y = income, by = id_area) A016 20 24.45873 5.319474 2.306399 9.429756
#> 4 mean(y = income, by = id_area) A025 20 25.18193 5.133222 2.265662 8.997172
#>           lower      upper
#> 1 10.82740 20.38334
#> 2 13.12284 21.40629
#> 3 19.93827 28.97919
#> 4 20.74132 29.62255
```

3 Case 2: Adding Non-Response

3.1 Context

We now introduce **unit non-response**. In the simulated data, approximately 60% of dwelling respond.

```
set.seed(123)
lfs_samp_dwel$rand <- runif(nrow(lfs_samp_dwel))
lfs_samp_dwel$rep <- lfs_samp_dwel$rand <= 0.6
lfs_samp_dwel$prep <- 0.6
```

Non-response is corrected by **reweighting**: the weights of respondents are adjusted to compensate for the non-respondents. Here, we use the constant probability of non-response. In practice, the probability of response is estimated by a logistic model. Be careful: use 0 instead of missing values for non-response weight. In these calculations, missing values create some trouble.

```
lfs_samp_dwel$w_final_nrc <- lfs_samp_dwel$w_final / lfs_samp_dwel$prep
lfs_samp_dwel$w_dwel_nrc  <- lfs_samp_dwel$w_dwel  / lfs_samp_dwel$prep

lfs_samp_dwel$w_final_nrc[!lfs_samp_dwel$rep] <- 0
lfs_samp_dwel$w_dwel_nrc[!lfs_samp_dwel$rep] <- 0
lfs_samp_dwel$income[!lfs_samp_dwel$rep] <- 0
```

3.2 Variance decomposition

When non-response is treated as a second phase of (Poisson) sampling, the total variance decomposes into two additive components (Caron, 1998):

$$V(\hat{Y}) = V_{\text{sampling}} + V_{\text{non-response}}$$

V_{sampling} is estimated using `var_srs()` on the full sample (respondents and non-respondents). Before computing the sampling variance, the variable of interest is divided by the response probability p_i , so that the sampling variance operates on the non-response-corrected variable. Non-respondents are reintroduced with zero values via `add_zero()`.

$$V_{\text{sampling}} = \sum_{h=1}^H N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}$$

where the sample variance within stratum h is computed on the corrected variable:

$$s_h^2 = \frac{1}{n_h - 1} \sum_{i \in h} \left(\frac{y_i}{p_i} - \bar{y}_h \right)^2$$

For non-respondents, $y_i = 0$, so they contribute to the variance through the deviation from the stratum mean.

$V_{\text{non-response}}$ is estimated using `var_pois()` on respondents only. It captures the additional variability introduced by the random nature of non-response:

$$V_{\text{non-response}} = \sum_{i \in \text{resp}} \frac{1}{\pi_i} (1 - p_i) \left(\frac{y_i}{p_i} \right)^2$$

where $\pi_i = n_h/N_h$ is the inclusion probability of unit i in stratum h , and p_i is the estimated response probability. In practice, p_i is estimated through logistic regression or within homogeneous response groups.

In **gustave**, this variance component is obtained with `var_pois(y, pik = p_i, w = 1/π_i)` where **pik** receives the response probabilities and **w** receives the sampling weights.

3.3 Using qvar()

`qvar()` integrates the non-response with two new parameters:

```
precision_nrc <- qvar(
  data      = lfs_samp_dwel,
  id        = "id_dwel",
  strata    = "id_area",
  dissemination_dummy = "rep",
  dissemination_weight = "w_final_nrc",
  sampling_weight      = "w_final",
  nrc_weight           = "w_final_nrc",
  response_dummy       = "rep",
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - stratified simple random sampling
#> - non-response correction through reweighting
#> Note: The strata variable (id_area) is of type character. It is automatically coerced
#>   ↪ to factor.
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

precision_nrc(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   ↪ discarded.
#>
#>           call      n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 524074026 22892.66 12.02035 145580.4 235318
```

Note the new arguments:

- `dissemination_dummy` now refers to `rep` (only respondents are disseminated).
- `nrc_weight`: the weight after non-response correction.
- `response_dummy`: indicates respondents.

Here we pass the whole `lfs_samp_dwel` dataset to the wrapper. `gustave` informs us that some units are discarded because of non-response. The result is the same with the subset of the dataset.

```
precision_nrc(lfs_samp_dwel %>% filter(rep == 1), total(income))
#>
#>           call      n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 524074026 22892.66 12.02035 145580.4 235318
```

3.4 Using define_variance_wrapper()

3.4.1 Variance function

The variance function now has two components: `var_pois` and `var_srs`. We also add some calculations for Poulpe comparison. We discuss them later.

```

diag_env <- new.env(parent = emptyenv())
var_sample_nrc <- function(y, lfs_samp_dwel) {

  variance <- list()

  resp <- lfs_samp_dwel %>% filter(rep == 1)

  # Component 1: Non-response variance (Poisson, respondents only)
  variance[["nr"]] <- var_pois(
    y[resp$id_dwel, ,drop = FALSE],
    pik = resp$prep,
    w = resp$w_final
  )

  # Poulpe calculations
  poulpe_F <- var_pois(
    y[resp$id_dwel, ,drop = FALSE] * resp$w_final,
    pik = resp$prep
  )

  # Reintroduce non-respondents with zeros
  y <- add_zero(y, rownames = lfs_samp_dwel$id_dwel)

  # Component 2: Sampling variance (SRS, full sample)
  # The variable must be divided by the response probability
  variance[["srs"]] <- var_srs(
    y / (lfs_samp_dwel$w_final / lfs_samp_dwel$w_final_nrc),
    pik = 1 / lfs_samp_dwel$w_final,
    strata = lfs_samp_dwel$id_area
  )

  diag_env$components <- variance
  diag_env$poulpe <- tibble(
    typeres = "FP", value = Reduce(`+`, variance) %>%
      add_row(typeres = "PF", value = NA) %>%
      add_row(typeres = "F", value = poulpe_F) %>%
      add_row(typeres = "E", value = variance[["nr"]] - poulpe_F) %>%
      add_row(typeres = "P", value = variance[["srs"]]) %>%
      add_row(typeres = "S", value = NA)
  )

  Reduce(`+`, variance)
}

```

`var_pois()` estimates the variance under Poisson sampling. It requires:

- `y`: the variable of interest (respondents only).
- `pik`: the response probabilities (estimated by $w_{\text{smp}}/w_{\text{nrc}}$).
- `w`: an optional weight vector (the sampling weight, to account for the first sampling stage).
- `drop = FALSE`: this option ensure that the row names indexation is kept during the subset

`add_zero()` is a key utility: it reintroduces the non-responding units into the `y` matrix by adding rows of zeros. This is necessary because `var_srs()` operates on the full sample.

A crucial point: the order matters.

1. **First**, estimate the non-response variance on respondents only.
2. **Then**, use `add_zero()` to expand `y` back to the full sample.
3. **Finally**, estimate the sampling variance on the full sample, after dividing by the response probability.

This ordering mirrors the mathematical decomposition: we work “inside out”, from the last correction step back to the first sampling step.

3.4.2 Technical data and wrapper creation

The creation of the wrapper is almost the same as for the case 1, except for some considerations about weights.

```
technical_data_nrc <- list()
technical_data_nrc$lfs_samp_dwel <- lfs_samp_dwel

precision2_nrc <- define_variance_wrapper(
  variance_function = var_sample_nrc,
  technical_data     = technical_data_nrc,
  reference_id       = technical_data_nrc$lfs_samp_dwel$id_dwel,
  reference_weight    = technical_data_nrc$lfs_samp_dwel$w_final_nrc,
  default_id         = "id_dwel"
)
```

Important notes:

- The `reference_weight` must be the **dissemination weight** (`wNrc`), not the sampling weight. Non-respondents have weight 0.
- NA values in `wNrc` must be replaced by 0 in `technical_data` before defining the wrapper.

3.4.3 Using the wrapper

When we use the wrapper, the data is automatically filtered to keep only responding units. A warning is shown to inform you. If the unit identifiers are ordered differently in the technical data and in the estimation data, `gustave` tells you that the rows have been reordered. This should have no impact on the variance estimation.

```
precision2_nrc(lfs_samp_dwel, total(income))
#>               call      n      est variance      std      cv      lower upper
#> 1 total(y = income) 80 190449.2 524074026 22892.66 12.02035 145580.4 235318
precision2_nrc(lfs_samp_dwel %>% filter(rep == 1), total(income))
#> Warning: Some observations from the survey appear to be missing. The variance
#>           ↪ estimation function may produce unexpected results.
#> Warning: The inputted id variable (id argument) appears not to match the reference id
#>           ↪ variable provided when the variance wrapper was defined: it is reordered and
#>           ↪ everything should be fine. Issues may nonetheless arise if part of the call is to be
#>           ↪ evaluated outside of the inputted data.frame (data argument).
#>               call      n      est variance      std      cv      lower upper
#> 1 total(y = income) 48 190449.2 524074026 22892.66 12.02035 145580.4 235318
```

Here we expect a higher variance in this case 2.

Table 1: Variance elements for Poulpe calculated from gustave

Component	types	gustave	value
Point estimate	S	est	190449.2
Sampling variance	P	‘var_srs()’ component	520860771.0
F-Non-response component	F		386107209.3
E-Non-response component	E		-382893954.5
Non-response variance	E+F	‘var_pois()’ component	3213254.8
Total variance	FP	var	524074025.8

```
precision2(lfs_samp_dwel, total(income))
#>      call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 114269.5 187509878 13693.42 11.98345 87430.87 141108.1
(results <- precision2_nrc(lfs_samp_dwel, total(income)))
#>      call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 524074026 22892.66 12.02035 145580.4 235318
```

3.5 Comparison with Poulpe

In our variance function, we reproduce all the calculation steps from Poulpe to compare with it. Here is the output from Poulpe :

**Table des résultats du calcul des estimateurs de totaux
et de variance issus des différents modules**

Obs	VAL1	types	passage	trdat	datexp	timexp	enqredrs	step
1	524074025.77	FP	0	POISSON	01SEP2026	15:40	BRUTES	SOMEPSV
2	209059999.91	PF	0	POISSONF	01SEP2026	15:40	BRUTES	DEFFPSV
3	386107209.27	F	0	POISSON	01SEP2026	15:40	BRUTES	FIFOPSV
4	-382893954.46	E	0	POISSON	01SEP2026	15:40	BRUTES	FIFOPSV
5	520860770.96	P	0	POISSON	01SEP2026	15:40	BRUTES	ESTIPSV
6	190449.20	S	0	POISSON	01SEP2026	15:40	BRUTES	ESTIPSV

In Poulpe, the non-response variance component is not displayed directly. Instead, Poulpe outputs two intermediate quantities, F and E, whose difference yields the non-response variance (see Caron, 1998, p. 185). Understanding the relationship between these quantities and the gustave formulation requires examining how the Poisson variance formula is parameterised.

The F component in Poulpe corresponds to a Poisson variance where the sampling weight has been absorbed into the variable of interest (i.e., the formula operates on $y_i/(\pi_i p_i)$ rather than on y_i with a separate weight argument):

$$V_{F\text{-poulpe}} = \sum_{i \in \text{resp}} (1 - p_i) \left(\frac{y_i}{p_i \pi_i} \right)^2$$

This is obtained in gustave by passing the pre-weighted variable to var_pois(): var_pois(y / pik_i, pik = p_i).

The **gustave** non-response variance, by contrast, keeps the sampling weight separate through the **w** argument:

$$V_{\text{NR}} = \sum_{i \in \text{resp}} \frac{1}{\pi_i} (1 - p_i) \left(\frac{y_i}{p_i} \right)^2$$

The difference between these two expressions is the E component:

$$V_{\text{E-poulpe}} = V_{\text{NR}} - V_{\text{F-poulpe}} = \sum_{i \in \text{resp}} (1 - p_i) \left(\frac{y_i}{p_i} \right)^2 \left(\frac{1}{\pi_i} - \frac{1}{\pi_i^2} \right)$$

The E-part of the Poulpe calculation is negative and can be obtained by setting $\text{var_pois}(y_i, p_i, 1/\pi - 1/\pi^2)$. This component is not a variance in itself, it is an intermediate result. The meaningful quantity is always $E + F$, which equals the **gustave** non-response component. In practice, verifying that the **gustave** total variance (FP) matches the **Poulpe** output is sufficient; the F and E components are useful for debugging but carry no independent methodological interpretation.

4 Case 3: Adding Calibration

4.1 Context

Here we account for **calibration** in the variance estimation. After non-response correction, the weights are typically calibrated on known population margins (e.g., totals by region, by sex, by age group) to reduce bias and improve precision.

Calibration reduces the variance of the estimator: by forcing the weighted sample to reproduce known population totals, it removes part of the variability.

4.2 Variance decomposition

There is no major change in the formula of variance estimation, we just need to replace the variable of interest **y** with **calibration residuals** before computing the variance components. These calibration residuals are obtained as the residuals of a weighted linear regression of **y** on the calibration variables **x**:

$$e_k = y_k - x'_k \hat{\beta}, \quad \text{where} \quad \hat{\beta} = (X'WX)^{-1} X'Wy$$

These residuals capture the part of **y** that is not explained by the calibration variables, which is the only part that contributes to the variance.

```
resp_dwel <- lfs_samp_dwel %>% filter(rep == 1)

# Example: if calibrating on id_area (catégorical)
e <- res_cal(resp_dwel$income, x = model.matrix(~ id_area, data = resp_dwel))
```

Important: `res_cal()` by default performs the regression **without an intercept**. The example above uses `model.matrix()` (intercept + treatment contrasts) instead. In `qvar()`, Qualitative calibration variables (character or factor) are automatically discretized into full sets of dummies.

For all the following cases, we assume the weights have already been calibrated on the variable **id_area**.

4.3 Using qvar()

```
lfs_samp_dwel <- as.data.frame(lfs_samp_dwel)

precision_cal <- qvar(
  data          = lfs_samp_dwel,
  id            = "id_dwel",
  strata        = "id_area",
  dissemination_dummy = "rep",
  dissemination_weight = "w_final_nrc",
  sampling_weight  = "w_final",
  nrc_weight       = "w_final_nrc",
  response_dummy   = "rep",
  calibration_dummy = "rep",
  calibration_weight = "w_final_nrc",
  calibration_var   = "id_area",
  define = TRUE
)
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#> - stratified simple random sampling
#> - non-response correction through reweighting
#> - calibration on margins
#> Note: The strata variable (id_area) is of type character. It is automatically coerced
  ↳ to factor.
#> Note: Note: The following calibration variables are qualitative (factor, character):
  ↳ id_area. They will be automatically discretized.
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

precision_cal(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
  ↳ discarded.
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 197245302 14044.4 7.374359 162922.6 217975.7
```

New arguments:

- `calibration_dummy`: which units were used in the calibration (typically, the respondents).
- `calibration_weight`: the calibrated weight.
- `calibration_var`: the calibration variable(s). For categorical variables, `qvar()` automatically creates the dummy matrix.

Note: `qvar()` does not accept tibbles when calibration variables are specified. Convert to `data.frame` first.

4.4 Using define_variance_wrapper()

4.4.1 Variance function

The variance function now includes a `res_cal()` step before the variance components:

```

diag_env <- new.env(parent = emptyenv())

var_sample_cal <- function(y, lfs_samp_dwel) {

  variance <- list()

  resp <- lfs_samp_dwel %>% filter(rep == 1)

  # Step 0: Prepare the calibration matrix (respondents only)
  cal_form <- as.formula("~id_area")
  x_cal <- model.matrix(cal_form, data = resp)

  # Step 1: Replace y with calibration residuals (respondents only)
  y[resp$id_dwel, ] <- res_cal(
    y[resp$id_dwel, ,drop = FALSE],
    x = x_cal,
    w = resp$w_final_nrc
  )

  # Step 2: Non-response variance (Poisson, respondents only)
  variance[["nr"]] <- var_pois(
    y[resp$id_dwel, ,drop = FALSE],
    pik = resp$prep,
    w = resp$w_final
  )

  poulpe_F <- var_pois(
    y[resp$id_dwel, ,drop = FALSE] * resp$w_final,
    pik = resp$prep
  )

  # Step 3: Reintroduce non-respondents with zeros
  y <- add_zero(y, rownames = lfs_samp_dwel$id_dwel)

  # Step 4: Sampling variance (SRS, full sample)
  variance[["srs"]] <- var_srs(
    y / (lfs_samp_dwel$prep),
    pik = 1 / lfs_samp_dwel$w_final,
    strata = lfs_samp_dwel$id_area
  )

  diag_env$components <- variance
  diag_env$poulpe <- tibble(typeres="FP",value=Reduce(`+`, variance)) %>%
    add_row(typeres="PF",value=NA) %>%
    add_row(typeres="F",value=poulpe_F) %>%
    add_row(typeres="E",value=variance[["nr"]] - poulpe_F) %>%
    add_row(typeres="P",value=variance[["srs"]]) %>%
    add_row(typeres="S",value=NA)

  Reduce(`+`, variance)
}

```

The key addition is **Step 1**: before any variance computation, we replace `y` with its calibration residuals. This reduces the variance because the part of `y` explained by the calibration variables is removed.

Here, we have hardcoded the calibration variables in the variance function itself. In practice, it is better to dynamically build the calibration matrix from all variables in `lfs_samp_dwel` used in the calibration model. We will see in the Practical Implementation section how to make the calibration variables fully configurable through the `technical_data` and `technical_param` mechanism, so that the same variance function can accommodate different calibration specifications without modifying its source code.

4.4.2 Technical data and wrapper creation

The creation of the wrapper is unchanged.

```
technical_data_cal <- list()
technical_data_cal$lfs_samp_dwel <- lfs_samp_dwel

precision2_cal <- define_variance_wrapper(
  variance_function = var_sample_cal,
  technical_data     = technical_data_cal,
  reference_id       = technical_data_cal$lfs_samp_dwel$id_dwel,
  reference_weight    = technical_data_cal$lfs_samp_dwel$w_final_nrc,
  default_id         = "id_dwel"
)
```

4.4.3 Using the wrapper

We expect a lower variance estimation with calibration step.

```
precision2(lfs_samp_dwel, total(income))
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 114269.5 187509878 13693.42 11.98345 87430.87 141108.1
precision_nrc(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
#>   discarded.
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 524074026 22892.66 12.02035 145580.4 235318
(results <- precision2_cal(lfs_samp_dwel, total(income)))
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 197245302 14044.4  7.374359 162922.6 217975.7
```

4.5 Comparison with Poulpe

Here is the variance calculated by Poulpe :

Table 2: Variance elements for Poulpe calculated from gustave

Component	typeres	gustave	value
Point estimate	S	est	190449.2
Sampling variance	P	‘var_srs()’ component	196618785.9
F-Non-response component	F		75341654.5
E-Non-response component	E		-74715138.6
Non-response variance	E+F	‘var_pois()’ component	626515.8
?	PF	?	NA
Total variance	FP	var	197245301.7

**Table des résultats du calcul des estimateurs de totaux
et de variance issus des différents modules**

Obs	VAL1	typeres	passage	trdat	datexp	timexp	enqredrs	step
1	197245301.70	FP	0	POISSON	01SEP2026	15:40	REDRES	SOMEPSV
2	187376386.60	PF	0	POISSONF	01SEP2026	15:40	REDRES	DEFFPSV
3	75341654.45	F	0	POISSON	01SEP2026	15:40	REDRES	FIFOPSV
4	-74715138.64	E	0	POISSON	01SEP2026	15:40	REDRES	FIFOPSV
5	196618785.88	P	0	POISSON	01SEP2026	15:40	REDRES	ESTIPSV
6	190449.20	S	0	POISSON	01SEP2026	15:40	REDRES	ESTIPSV

5 Case 4: Two-Stage Sampling

5.1 Context

In many dwelling surveys at Statbel (such as LFS), the sampling design involves two stages. First, a random sample of municipalities is drawn from the population. Then a stratified random sample of dwellings is drawn. Some of them do not respond, and answers are at dwelling level. At last, weights are corrected through reweighting.

5.2 Variance decomposition

This two-stage design introduces an additional source of variability compared to the previous cases. The total variance now decomposes into three additive components:

$$V(\hat{Y}) = V_{PSU} + V_{HH} + V_{NR}$$

The first component captures the uncertainty from drawing PSUs from the population. The second captures the uncertainty from drawing dwellings from the selected PSU. The last component accounts for the random nature of dwelling non-response.

Let us define the following quantities. At the PSU level: N_P is the total number of PSUs in the population, n_P is the number of sampled PSUs, and π_{psu} is the first-order inclusion probability of a PSU. At the dwelling level: $N_{h|j}$ is the number of dwellings in PSU j for the strata h , $n_{h|j}$ is the number of sampled dwellings in

PSU j for the strata h , and $\pi_{h|j}$ is the conditional inclusion probability of dwelling h given that PSU j and strata h has been selected. The overall inclusion probability of a dwelling is $\pi_h = \pi_{\text{psu}(h)} \cdot \pi_{h|\text{psu}(h)}$. Finally, p_h denotes the estimated response probability for dwelling h .

Non-response is modeled as a Poisson sampling phase. For each responding dwelling, the contribution to the non-response variance depends on its response probability and its fully expanded value. The formula is:

$$V_{\text{NR}} = \sum_{i \in \text{resp}} \frac{1}{\pi_i} (1 - p_i) \left(\frac{y_i}{p_i} \right)^2$$

Within each selected PSU, a simple random sample of dwellings is drawn. The sub-sampling variance is the sum, across all sampled PSUs, of the within-PSU SRS variance, each weighted by the inverse of the PSU inclusion probability to account for the fact that only a fraction of the PSUs are observed:

$$V_{HH} = \sum_{j=1}^{n_P} \frac{1}{\pi_{\text{psu},j}} \left[\sum_{h=1}^H N_{h|j}^2 \left(1 - \frac{n_{h|j}}{N_{h|j}} \right) \frac{s_{h|j}^2}{n_{h|j}} \right]$$

where $s_{h|j}^2$ is the sample variance of the non-response-corrected variable y_h/p_h within PSU j . In the R code, this is implemented by looping over PSUs and calling `var_srs()` on each subset, then multiplying by $1/\pi_{\text{psu}}$.

Before computing the PSU-level variance, the dwelling-level variable must be aggregated to PSU totals. For each PSU j , the estimated PSU total is obtained by expanding the non-response-corrected dwelling values using the within-PSU Horvitz-Thompson estimator:

$$\hat{Y}_j = \sum_{h \in j} \frac{1}{\pi_{h|j}} \frac{y_h}{p_h}$$

In `gustave`, this aggregation is performed by `sum_by()` with the weight argument set to $1/\pi_{h|j}$.

Once the PSU-level totals $\hat{Y}_j$ are available, the PSU sampling variance follows the standard stratified SRS formula:

$$V_{PSU} = N_P^2 \left(1 - \frac{n_P}{N_P} \right) \frac{s_P^2}{n_P}$$

Where s_P^2 is the sample variance of the $\hat{Y}_j$ values. In `gustave`, this is a single call to `var_srs()`.

This case cannot be handled by `qvar()`, which is limited to single-stage designs. It is precisely the kind of situation where `define_variance_wrapper()` becomes essential: we need to write a custom variance function that chains the three components in the correct order — from the innermost stage (dwelling sub-sampling) back to the outermost stage (PSU sampling), applying `add_zero()` at each transition to reintroduce the units lost at the previous step.

We explain step by step the `define_variance_wrapper()` writing, then we introduce a custom `qvar_ms()` function, to estimate variance for multi-levels stratified simple random sampling.

For this case, we use the two datasets `lfs_samp_area` and `lfs_samp_dwel`.

5.3 Using `define_variance_wrapper()`

5.3.1 Variance function

```

diag_env <- new.env(parent = emptyenv())
var_sample_two <- function(y, lfs_samp_area, lfs_samp_dwel) {

  variance <- list()

  # Step 0: Prepare the respondents datasets
  resp_dwel <- lfs_samp_dwel %>% filter(rep == 1)

  # Step 1: Prepare the calibration matrix (individual level)
  cal_form <- as.formula("~id_area")
  x_cal <- model.matrix(cal_form, data = resp_dwel)

  # Step 2: Replace y with calibration residuals
  y[resp_dwel$id_dwel, ] <- res_cal(
    y[resp_dwel$id_dwel,],
    x = x_cal,
    w = resp_dwel$w_final_nrc
  )

  # Step 3: Non-response variance (Poisson, individuals respondents only)
  variance[["nr"]] <- var_pois(
    y[resp_dwel$id_dwel, ,drop = FALSE],
    pik = resp_dwel$prep,
    w = resp_dwel$w_final
  )

  poulpe_F <- var_pois(
    y[resp_dwel$id_dwel, ,drop = FALSE] * resp_dwel$w_final,
    pik = resp_dwel$prep
  )

  # Step 4: Reintroduce dwellings non-respondents with zeros
  y <- add_zero(y, rownames = lfs_samp_dwel$id_dwel)

  # Step 5: One variance per PSU, weighted by PSU size
  variance[["srs-dwel"]] <- unique(lfs_samp_dwel$id_area) %>% map(~{
    sub <- lfs_samp_dwel %>% filter(id_area == .x)
    pik_area <- 1 / unique(sub$w_area)
    var_srs(
      y[sub$id_dwel, ,drop = FALSE] / sub$prep,
      pik = 1 / sub$w_dwel
    ) * 1/pik_area
  }) %>% unlist %>% sum

  # --- Transition: aggregate to area level ---
  y_area <- sum_by(y / lfs_samp_dwel$prep,
    by = lfs_samp_dwel$id_area,
    w = lfs_samp_dwel$w_dwel)

  # Step 6: area sampling variance
  variance[["srs-area"]] <- var_srs(
    y_area[lfs_samp_area$id_area, , drop = FALSE],
    pik = 1 / lfs_samp_area$w_area
  )

```

```

diag_env$components <- variance
diag_env$poulpe <- tibble(type="FP",value=Reduce(`+`, variance)) %>%
  add_row(type="PF",value=NA) %>%
  add_row(type="F",value=poulpe_F) %>%
  add_row(type="E",value=variance[["nr"]] - poulpe_F) %>%
  add_row(type="P",value=variance[["srs-dwel"]] + variance[["srs-area"]]) %>%
  add_row(type="S",value=NA)

Reduce(`+`, variance)
}

```

The key addition is two aggregations, one at dwelling then at area level. The calibration is done at individual level. There are two `var_srs()` steps, the key step here is to estimate variance of dwelling within each area, and then to divide by the selection probability of each area. After that, we aggregate from dwelling to area with the non-response correction and finally calculate the variance due to the area sampling.

5.3.2 Technical data and Wrapper

```

technical_data_two <- list()
technical_data_two$lfs_samp_area <- lfs_samp_area
technical_data_two$lfs_samp_dwel <- lfs_samp_dwel

precision_two <- define_variance_wrapper(
  variance_function = var_sample_two,
  technical_data = technical_data_two,
  reference_id = technical_data_two$lfs_samp_dwel$id_dwel,
  reference_weight = technical_data_two$lfs_samp_dwel$w_final_nrc,
  default_id = "id_dwel"
)

```

5.3.3 Using the Wrapper

```

(results <- precision_two(lfs_samp_dwel %>% filter(rep == 1), total(income)))
#> Warning: Some observations from the survey appear to be missing. The variance
#>   ↪ estimation function may produce unexpected results.
#> Warning: The inputted id variable (id argument) appears not to match the reference id
#>   ↪ variable provided when the variance wrapper was defined: it is reordered and
#>   ↪ everything should be fine. Issues may nonetheless arise if part of the call is to be
#>   ↪ evaluated outside of the inputted data.frame (data argument).
#>
#>      call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2  8933292 2988.861 1.569375 184591.1 196307.2

```

5.4 Using `qvar_ms()`

`qvar_ms()` extends `qvar()` to multistage sampling designs, that is: stratified simple random sampling at each stage (units of stage k being drawn within the units of stage $k - 1$) ; non-response correction (if any) through reweighting, possibly at each stage ; calibration (if any) at one given stage.

To use `qvar_ms()`, you need, at each stage of sampling, the conditional sampling weight, the probability of response (if any) and the calibration weight (if any). Parameters are quite identical as `qvar()`, except you need list to declare argument for each stage.

```
precision2_two <- qvar_ms(

  # One file per stage, from the highest to the lowest
  data = list(lfs_samp_area, lfs_samp_dwel),
  sampling_stages = 2,
  id = c("id_area", "id_dwel"),
  parent_id = "id_area",

  # Dissemination and scope (last stage only)
  dissemination_dummy = "rep",
  dissemination_weight = "w_final_nrc",

  # Sampling design (conditional weights)
  sampling_weight = c("w_area", "w_dwel"),

  # Non-response correction (last stage by default)
  response_prob = "prep",
  response_dummy = "rep",

  # Calibration (last stage by default, cumulated weight)
  calibration_weight = "w_final_nrc",
  calibration_var = "id_area",

  define = TRUE
)
#> Note: The response_prob argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 2).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 2-stage sampling design
#>   - stage 1: simple random sampling WITHOUT stratification
#>   - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#>   - calibration on margins at stage 2
#> Note: The following calibration variables are qualitative (factor, character):
  ↳ id_area. They will be automatically discretized.
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

precision2_two(lfs_samp_dwel, total(income))
#> Warning: 32 observations do not match any responding units of the survey. They are
  ↳ discarded.
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2  8933292 2988.861 1.569375 184591.1 196307.2
```

Table 3: Variance elements for Poulpe calculated from gustave

Component	types	gustave	value
Point estimate	S	est	190449.2
Sampling variance	P	‘var_srs()’ component	8306775.9
F-Non-response component	F		75341654.5
E-Non-response component	E		-74715138.6
Non-response variance	E+F	‘var_pois()’ component	626515.8
?	PF	?	NA
Total variance	FP	var	8933291.8

5.5 Comparison with Poulpe

Here is the variance calculated by Poulpe :

**Table des résultats du calcul des estimateurs de totaux
et de variance issus des différents modules**

Obs	VAL1	types	passage	trdat	datexp	timexp	enqredrs	step
1	8933291.29	FP	0	POISSON	01SEP2026	15:40	REDRES	SOMEPSV
2	187376366.99	PF	0	POISSONF	01SEP2026	15:40	REDRES	DEFFPSV
3	75341646.58	F	0	POISSON	01SEP2026	15:40	REDRES	FIFOPSV
4	-74715130.80	E	0	POISSON	01SEP2026	15:40	REDRES	FIFOPSV
5	8306775.51	P	0	POISSON	01SEP2026	15:40	REDRES	ESTIPSV
6	190449.10	S	0	POISSON	01SEP2026	15:40	REDRES	ESTIPSV

6 Case 5: Adding individuals answers

6.1 Context

Case 4 assumed that the variable of interest was measured at the dwelling level. In practice, however, most survey statistics are produced at the individual level — for instance, unemployment rates or average income per person. In this case, the sampling design remains a two-stage draw (area sampling, then dwelling SRS within areas), followed by an exhaustive collection of all individuals within responding dwellings. The third sampling stage is exhaustive : once a dwelling responds, all its members are observed.

Once again, `qvar()` cannot estimate variance for a three-stage sampling design. We need to develop a variance wrapper by ourself. `qvar_ms()` integrate natively this case. As we see in this section, the introduction of the individual ask some questions about which stage the non-response and the calibration occur. We explain the consequences of each choice, present the choice made by Poulpe and how to use `qvar_ms()` here.

6.2 Variance decomposition

The total variance still decomposes into the same three additive components as in Case 4:

$$V(\hat{Y}) = V_{PSU} + V_{HH} + V_{NR}$$

The formulas are identical to those presented in Case 4. The only difference is operational: the variable of interest y_i is now defined at the individual level and must be aggregated to the dwelling level (by summing within dwellings) before computing the dwelling and area variance components. This aggregation step occurs naturally through `sum_by()`.

6.2.1 Non-response level: dwelling or individual?

A subtle but important question arises: at which level should non-response be modeled? In reality, non-response occurs at the dwelling level — when a dwelling refuses to participate, all its members are lost simultaneously. However, in the **Poulpe** methodology, non-response is always treated as the last phase of the sampling design, which means it is applied at the level of the analysis unit — here, the individual.

These two approaches lead to different variance estimates because they imply different models for the non-response mechanism. When non-response is modeled at the dwelling level, the response probability p_h is the same for all members of dwelling h , and the non-response variance is computed on dwelling-level aggregates. When non-response is modeled at the individual level, each individual carries the response probability of their dwelling, but the Poisson variance formula is applied to individual-level values — which implicitly treats each individual as an independent response unit.

We implement both approaches below. The first version (`var_sample_three_1()`) models non-response at the individual level, consistent with the **Poulpe** convention. The second version (`var_sample_three_2()`) models non-response at the dwelling level, which better reflects the actual data-generating process. Comparing the two helps assess the sensitivity of the variance estimate to this modelling choice and validates the R implementation against **Poulpe** reference values.

6.2.2 Calibration level

The level at which calibration residuals are computed depends on the non-response modelling choice.

In the **Poulpe**-consistent approach (version 1), calibration is applied at the individual level before any aggregation. The calibration matrix contains individual-level variables alongside dwelling-level variables that are simply repeated for each dwelling member. This is consistent with **Poulpe**'s convention of applying a single `res_cal()` at the analysis unit level.

In the dwelling-level approach (version 2), the individual variable of interest is first aggregated to the dwelling level. Calibration is then applied at the dwelling level. If the actual calibration used both dwelling margins and individual margins, the individual calibration variables must be aggregated to the dwelling level — typically by summing the dummy indicators within each dwelling — before being included in the `res_cal()` calibration matrix. This produces a dwelling-level residual that accounts for both types of margins simultaneously.

For this case, we use the three datasets `lfs_samp_area`, `lfs_samp_dwel` and `lfs_samp_ind`. We need to make some changes to the datasets before using them. In particular, we copy the `pik_dwel` into the individual dataset.

```
lfs_samp_ind <- lfs_samp_ind %>%
  select(-any_of(c("pik_area", "w_area", "w_dwel", "w_final", "w_final_nrc",
                  "prep", "rep", "id_area")))) %>%
  left_join(lfs_samp_dwel %>%
    select(id_dwel, id_area, rep, prep, w_dwel, w_final,
           w_final_nrc), by = "id_dwel") %>%
  left_join(lfs_samp_area %>%
    select(id_area, pik_area, w_area), by = "id_area")
```

6.3 Using define_variance_wrapper()

6.3.1 Variance function

In `var_sample_three_1()`, `res_cal()` and `var_pois()` are done before dwelling aggregation.

```
diag_env_1 <- new.env(parent = emptyenv())
var_sample_three_1 <- function(y, lfs_samp_area, lfs_samp_dwel, lfs_samp_ind) {

  variance <- list()
  poulpe <- list()

  # Step 0: Prepare the respondents datasets
  resp_ind <- lfs_samp_ind %>% filter(rep == 1)
  resp_dwel <- lfs_samp_dwel %>% filter(rep == 1)
  y <- y[resp_ind$id_ind, , drop = FALSE]

  # Step 1: Prepare the calibration matrix (individual level)
  cal_form <- as.formula("~id_area")
  x_cal <- model.matrix(cal_form, data = resp_ind)

  # Step 2: Replace y with calibration residuals
  y[resp_ind$id_ind, ] <- res_cal(
    y[resp_ind$id_ind, ],
    x = x_cal,
    w = resp_ind$w_final_nrc
  )

  # Step 3: Non-response variance (Poisson, individuals respondents only)
  variance[["nr"]] <- var_pois(
    y[resp_ind$id_ind, , drop = FALSE],
    pik = resp_ind$prep,
    w = resp_ind$w_final
  )

  poulpe_F <- var_pois(
    y[resp_ind$id_ind, , drop = FALSE] * resp_ind$w_final,
    pik = resp_ind$prep
  )

  # Step 4: Reintroduce individuals non-respondents with zeros
  y <- add_zero(y, rownames = lfs_samp_ind$id_ind)

  # --- Transition: aggregate to dwelling level ---
  y_dwel <- sum_by(y / lfs_samp_ind$prep, by = lfs_samp_ind$id_dwel)

  # Step 5: Reintroduce dwellings non-respondents with zeros
  y_dwel <- add_zero(y_dwel, rownames = lfs_samp_dwel$id_dwel)

  # Step 6: One variance per area, weighted by area size
  variance[["srs-dwel"]] <- unique(lfs_samp_dwel$id_area) %>% map(~{
    sub <- lfs_samp_dwel %>% filter(id_area == .x)
    pik_area <- 1 / unique(sub$w_area)
    var_srs(
      y_dwel[sub$id_dwel, , drop = FALSE],
```

```

    pik = 1 / sub$w_dwel
  ) * 1/pik_area
}) %>% unlist %>% sum

# --- Transition: aggregate to area level ---
y_area <- sum_by(y_dwel,
  by = lfs_samp_dwel$id_area,
  w = lfs_samp_dwel$w_final)

# Step 7: area sampling variance
variance[["srs-area"]] <- var_srs(
  y_area[lfs_samp_area$id_area, , drop = FALSE],
  pik = 1 / lfs_samp_area$w_area
)

diag_env_1$components <- variance
diag_env_1$poulpe <- tibble(ypes="FP",value=Reduce(`+`, variance)) %>%
  add_row(ypes="PF",value=NA) %>%
  add_row(ypes="F",value=poulpe_F) %>%
  add_row(ypes="E",value=variance[["nr"]] - poulpe_F) %>%
  add_row(ypes="P",value=variance[["srs-area"]] + variance[["srs-dwel"]]) %>%
  add_row(ypes="S",value=NA)

Reduce(`+`, variance)
}

```

In `var_sample_three_2()`, `res_cal()` and `var_pois()` are done after dwelling aggregation.

```

diag_env_2 <- new.env(parent = emptyenv())
var_sample_three_2 <- function(y, lfs_samp_area, lfs_samp_dwel, lfs_samp_ind) {

  variance <- list()
  poulpe <- list()

  # Step 0: Prepare the respondents datasets
  resp_ind <- lfs_samp_ind %>% filter(rep == 1)
  resp_dwel <- lfs_samp_dwel %>% filter(rep == 1)
  y <- y[resp_ind$id_ind, , drop = FALSE]

  # --- Transition: aggregate to dwelling level ---
  y_dwel <- sum_by(y, by = resp_ind$id_dwel)

  # Step 1: Prepare the calibration matrix (dwelling level)
  cal_form <- as.formula("~id_area")
  x_cal <- model.matrix(cal_form, data = resp_dwel)

  # Step 2: Replace y with calibration residuals
  y_dwel[resp_dwel$id_dwel, ] <- res_cal(
    y_dwel[resp_dwel$id_dwel,],
    x = x_cal,
    w = resp_dwel$w_final_nrc
  )

  # Step 3: Non-response variance (Poisson, dwelling respondents)

```

```

variance[["nr"]] <- var_pois(
  y_dwel[resp_dwel$id_dwel, ,drop = FALSE],
  pik = resp_dwel$prep,
  w = resp_dwel$w_final
)

poulpe_F <- var_pois(
  y_dwel[resp_dwel$id_dwel, ,drop = FALSE] * resp_dwel$w_final,
  pik = resp_dwel$prep
)

# Step 4: Reintroduce dwellings non-respondents with zeros
y_dwel <- add_zero(y_dwel, rownames = lfs_samp_dwel$id_dwel)

# Step 5: One variance per area, weighted by area size
variance[["srs-dwel"]] <- unique(lfs_samp_dwel$id_area) %>% map(~{
  sub <- lfs_samp_dwel %>% filter(id_area == .x)
  pik_area <- 1 / unique(sub$w_area)
  var_srs(
    y_dwel[sub$id_dwel, ,drop = FALSE] / sub$prep,
    pik = 1 / sub$w_dwel
  ) * 1/pik_area
}) %>% unlist %>% sum

# --- Transition: aggregate to PSU level ---
y_area <- sum_by(y_dwel / lfs_samp_dwel$prep,
  by = lfs_samp_dwel$id_area,
  w = lfs_samp_dwel$w_dwel)

# Step 6: area sampling variance
variance[["srs-area"]] <- var_srs(
  y_area[lfs_samp_area$id_area, , drop = FALSE],
  pik = 1 / lfs_samp_area$w_area
)

diag_env_2$components <- variance
diag_env_2$poulpe <- tibble(ypes="FP",value=Reduce(`+`, variance)) %>%
  add_row(ypes="PF",value=NA) %>%
  add_row(ypes="F",value=poulpe_F) %>%
  add_row(ypes="E",value=variance[["nr"]] - poulpe_F) %>%
  add_row(ypes="P",value=variance[["srs-area"]] + variance[["srs-dwel"]]) %>%
  add_row(ypes="S",value=NA)

Reduce(`+`, variance)
}

```

6.3.2 Technical data and Wrapper

```

technical_data_three <- list()
technical_data_three$lfs_samp_area <- lfs_samp_area
technical_data_three$lfs_samp_dwel <- lfs_samp_dwel
technical_data_three$lfs_samp_ind <- lfs_samp_ind

```

```
precision_three_1 <- define_variance_wrapper(
  variance_function = var_sample_three_1,
  technical_data    = technical_data_three,
  reference_id      = technical_data_three$lfs_samp_ind$id_ind,
  reference_weight  = technical_data_three$lfs_samp_ind$w_final_nrc,
  default_id       = "id_ind"
)

precision_three_2 <- define_variance_wrapper(
  variance_function = var_sample_three_2,
  technical_data    = technical_data_three,
  reference_id      = technical_data_three$lfs_samp_ind$id_ind,
  reference_weight  = technical_data_three$lfs_samp_ind$w_final_nrc,
  default_id       = "id_ind"
)
```

6.3.3 Using the Wrapper

```
(results_1 <- precision_three_1(lfs_samp_ind %>% filter(rep == 1), total(income)))
#> Warning: Some observations from the survey appear to be missing. The variance
  ↳ estimation function may produce unexpected results.
#> Warning: The inputted id variable (id argument) appears not to match the reference id
  ↳ variable provided when the variance wrapper was defined: it is reordered and
  ↳ everything should be fine. Issues may nonetheless arise if part of the call is to be
  ↳ evaluated outside of the inputted data.frame (data argument).
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2  7242261 2691.145 1.413052 185174.6 195723.7
(results_2 <- precision_three_2(lfs_samp_ind %>% filter(rep == 1), total(income)))
#> Warning: Some observations from the survey appear to be missing. The variance
  ↳ estimation function may produce unexpected results.
#> Warning: The inputted id variable (id argument) appears not to match the reference id
  ↳ variable provided when the variance wrapper was defined: it is reordered and
  ↳ everything should be fine. Issues may nonetheless arise if part of the call is to be
  ↳ evaluated outside of the inputted data.frame (data argument).
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2  8933292 2988.861 1.569375 184591.1 196307.2
```

6.4 Using qvar_ms()

The two variance functions above are about fifty lines each, and almost all of that code is plumbing: selecting respondents, expanding by the response probability, aggregating with `sum_by()`, reintroducing zeros with `add_zero()`, looping over areas, reindexing before each `var_srs()`. Only two lines actually carry the methodological choice — where `res_cal()` and `var_pois()` are placed relative to the dwelling aggregation. Everything else is the same in both versions, and would be the same again for any other three-stage design.

`qvar_ms()` reverses this ratio. The design is declared once, argument by argument, and the pipeline is built from the declaration. The two models then differ by two arguments only: in model 1, `response_prob` and the calibration are declared at the last (individual) stage; in model 2, they are declared at the dwelling stage, through `response_prob = list(NULL, "prep", NULL)` and `calibration_stage = 2`. Nothing else changes.

Two points deserve attention in the calls below. First, the individual stage is exhaustive: all members of a responding dwelling are surveyed, so `w_ind = 1` and the third-stage sampling variance is zero. The stage is still declared, because it is where the analysis units live. Second, `sampling_weight` expects conditional weights — the weight of a unit given that its parent was selected — whereas `calibration_weight` expects the cumulated weight actually used in the calibration equations. This is the most common source of error with `qvar_ms()`, and the dedicated vignette comes back to it at length.

```
lfs_samp_ind$w_ind <- 1

precision2_three_1 <- qvar_ms(

  # One file per stage, from the highest to the lowest
  data = list(lfs_samp_area, lfs_samp_dwel, lfs_samp_ind),
  sampling_stages = 3,
  id = c("id_area", "id_dwel", "id_ind"),
  parent_id = c("id_area", "id_dwel"),

  # Dissemination and scope (last stage only)
  dissemination_dummy = "rep",
  dissemination_weight = "w_final_nrc",

  # Sampling design (conditional weights)
  sampling_weight = c("w_area", "w_dwel", "w_ind"),

  # Non-response correction (last stage by default)
  response_prob = "prep",
  response_dummy = "rep",

  # Calibration (last stage by default, cumulated weight)
  calibration_weight = "w_final_nrc",
  calibration_var = "id_area",

  define = TRUE
)

#> Note: The response_prob argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 3).
#> Note: The response_dummy argument is of length 1: it is assumed to refer to the last
  ↳ sampling stage (stage 3).
#> Survey variance estimation with the gustave package
#>
#> The following features are taken into account:
#>   - 3-stage sampling design
#>   - stage 1: simple random sampling WITHOUT stratification
#>   - stage 2: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above)
#>   - stage 3: simple random sampling WITHOUT stratification (within the units of the
  ↳ stage above), with non-response correction through reweighting
#>   - calibration on margins at stage 3
#> Note: The following calibration variables are qualitative (factor, character):
  ↳ id_area. They will be automatically discretized.
#> Note: At stage 2, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_area).
#> Note: At stage 3, units are drawn within the units of the stage above: the effective
  ↳ strata used for variance estimation are the crossing of the parent unit (id_dwel).
#> Warning: The following strata at stage 3 contain less than two sampled units:
  ↳ D00165:1, D00174:1, D00238:1, D00281:1, D00387:1, D00630:1, D00637:1, D00784:1,
  ↳ D01391:1, D01784:1 and 37 more. They are excluded from the stratified SRS variance
  ↳ estimation (but kept for point estimates)31
```

```
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.
```

```
precision2_three_1(lfs_samp_ind,total(income))
```

```
#> Warning: 48 observations do not match any responding units of the survey. They are  
↪ discarded.
```

```
#>      call      n      est variance      std      cv      lower      upper  
#> 1 total(y = income) 68 190449.2 7242261 2691.145 1.413052 185174.6 195723.7
```

```
lfs_samp_ind$w_ind <- 1
```

```
precision2_three_2 <- qvar_ms(
```

```
  # One file per stage, from the highest to the lowest
```

```
  data = list(lfs_samp_area, lfs_samp_dwel,  
              lfs_samp_ind %>% filter(rep)),
```

```
  sampling_stages = 3,
```

```
  id = c("id_area", "id_dwel", "id_ind"),
```

```
  parent_id = c("id_area", "id_dwel"),
```

```
  # Dissemination and scope (last stage only)
```

```
  dissemination_dummy = "rep",
```

```
  dissemination_weight = "w_final_nrc",
```

```
  # Sampling design (conditional weights)
```

```
  sampling_weight = c("w_area", "w_dwel", "w_ind"),
```

```
  # Non-response correction (last stage by default)
```

```
  response_prob = list(NULL,"prep",NULL),
```

```
  response_dummy = list(NULL,"rep",NULL),
```

```
  # Calibration (last stage by default, cumulated weight)
```

```
  calibration_weight = "w_final_nrc",
```

```
  calibration_var = "id_area",
```

```
  calibration_stage = 2,
```

```
  define = TRUE
```

```
)
```

```
#> Survey variance estimation with the gustave package
```

```
#>
```

```
#> The following features are taken into account:
```

```
#> - 3-stage sampling design
```

```
#> - stage 1: simple random sampling WITHOUT stratification
```

```
#> - stage 2: simple random sampling WITHOUT stratification (within the units of the  
↪ stage above), with non-response correction through reweighting
```

```
#> - stage 3: simple random sampling WITHOUT stratification (within the units of the  
↪ stage above)
```

```
#> - calibration on margins at stage 2
```

```
#> Note: The following calibration variables are qualitative (factor, character):
```

```
↪ id_area. They will be automatically discretized.
```

```
#> Note: At stage 2, units are drawn within the units of the stage above: the effective  
↪ strata used for variance estimation are the crossing of the parent unit (id_area).
```

```
#> Note: At stage 3, units are drawn within the units of the stage above: the effective  
↪ strata used for variance estimation are the crossing of the parent unit (id_dwel).
```

```
#> Warning: The following strata at stage 3 contain less than two sampled units:
```

```
↪ D00165:1, D00281:1, D00387:1, D00637:1, D00784:1, D01391:1, D02523:1, D02574:1,
```

```
↪ D02653:1, D03338:1 and 18 more. They are excluded from the stratified SRS variance
```

```
↪ estimation (but kept for point estimates)32
```

```
#> Note: As define = TRUE, a ready-to-use variance wrapper is (invisibly) returned.

precision2_three_2(lfs_samp_ind,total(income))
#> Warning: 48 observations do not match any responding units of the survey. They are
  ↪ discarded.
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2  8933292 2988.861 1.569375 184591.1 196307.2
```

6.5 Comparing the two models

The point estimate is identical in both versions (190 449.2), as it must be: the two wrappers encode the same final weights, and only the *description* of how those weights were built differs — non-response and calibration declared at the individual stage (model 1) or at the dwelling stage (model 2). Everything that follows is read directly from the two `diagnostics = "comp"` printouts above (see the dedicated section below for their general use).

```
precision2_three_1(lfs_samp_ind,total(income), diagnostics = "comp")
#> Warning: 48 observations do not match any responding units of the survey. They are
  ↪ discarded.
#>
#> -- Variance components: calibration effect --
#>           with calib.      share           without      effect
#> nr_stage3      5.986e+05    8.3 % #      2.436e+06    -75.4 %
#> sampling_stage3      0      0.0 %           0      NaN %
#> sampling_stage2      6.644e+06  91.7 % #####      2.2e+07    -69.8 %
#> sampling_stage1      3.68e-22    0.0 %           3.271e+08   -100.0 %
#> TOTAL      7.242e+06  100.0 % #####      3.515e+08    -97.9 %
#>
#> Calibration multiplies the variance by 0.021 (deft x 0.144).
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2  7242261 2691.145 1.413052 185174.6 195723.7
precision2_three_2(lfs_samp_ind,total(income), diagnostics = "comp")
#> Warning: 48 observations do not match any responding units of the survey. They are
  ↪ discarded.
#>
#> -- Variance components: calibration effect --
#>           with calib.      share           without      effect
#> sampling_stage3      0      0.0 %           0      NaN %
#> nr_stage2      6.265e+05    7.0 % #      3.213e+06    -80.5 %
#> sampling_stage2      8.307e+06  93.0 % #####      2.2e+07    -62.2 %
#> sampling_stage1      7.031e-23    0.0 %           3.271e+08   -100.0 %
#> TOTAL      8.933e+06  100.0 % #####      3.523e+08    -97.5 %
#>
#> Calibration multiplies the variance by 0.025 (deft x 0.159).
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 68 190449.2  8933292 2988.861 1.569375 184591.1 196307.2
```

Start with the “without calibration” column: the two models are practically equivalent (5.515×10^8 vs 5.523×10^8). Since the response probability is constant within each dwelling, expanding the individual values by $1/\hat{p}$ and then aggregating to dwelling totals (model 1) gives the same totals as aggregating first and expanding afterwards (model 2): the sampling components of stages 1 and 2 are therefore identical by construction (2.271×10^8 and 2.2×10^7 in both printouts). Only the non-response component itself differs:

2.436×10^6 at the individual stage versus 3.213×10^6 at the dwelling stage (+32%). The dwelling-level Poisson term operates on aggregated totals — larger squared values — summed over fewer units, which reduces the averaging effect. But this component weighs less than 0.6% of the uncalibrated total: for this design, *where the non-response correction is declared is almost immaterial to the uncalibrated variance*.

Calibration is the real lever — and its position is what separates the models. The calibration variable is `id_area`: a post-stratification on the primary sampling units themselves. The estimated area totals are thereby constrained to fixed margins, so their sampling variability vanishes: the `sampling_stage1` component collapses to zero in both models. Since stage 2 carried 97.7% of the uncalibrated variance, the overall effect is dramatic — the variance is decreased by 97.9% (model 1) and 97.5% (model 2). This is a deliberately extreme, didactic margin: calibrating on the PSUs amounts to stratifying *ex post* on them, and real-life margins (socio-demographic totals) will reduce, not annihilate, the first-stage component.

The remaining 24% gap between the models (7.242×10^6 vs 8.933×10^6 , CV 1.41% vs 1.57%) is entirely a matter of where the residualization occurs. In model 2, the dwelling totals are replaced by their calibration residuals at stage 2, *before* the stage-2 non-response and sampling terms are computed: calibration then absorbs 80.5% of the non-response component and 62.2% of the stage-2 sampling component. In model 1, the residuals are taken at the individual stage and only reach stage 2 through aggregation, absorbing less (−75.4% and −69.8%). Residualizing at the stage where the variance actually lives captures more of it — a general lesson worth remembering when a survey leaves the calibration stage open.

In practice, the declaration should follow the actual mechanisms: in this survey, non-response occurs at the dwelling level and the calibrated weights are dwelling-level weights, so model 2 is the faithful description — and, incidentally, the more efficient one. Model 1 remains the right choice when the goal is to replicate an existing *Poulpe* run specified at the individual level.

6.6 Comparison with Poulpe

The comparison with *Poulpe* can be done only with the first situation, with a correction at individual level. Two small differences appear in the *Poulpe* output. First, the point estimate returned under `typeres = "S"` (191258.50) is not the weighted total of the variable of interest (190449.16), obtained directly from the dissemination weights). Second, the total variance (7273036.7) is slightly above the one estimated in R.

Table des résultats du calcul des estimateurs de totaux et de variance issus des différents modules

Obs	VAL1	typeres	passage	trdat	datexp	timexp	enqredrs	step
1	7273036.68	FP	0	POISSON	02SEP2026	12:43	REDRES	SOMEPSV
2	180497443.62	PF	0	POISSONF	02SEP2026	12:43	REDRES	DEFFPSV
3	72598572.43	F	0	POISSON	02SEP2026	12:43	REDRES	FIFOPSV
4	-71997415.03	E	0	POISSON	02SEP2026	12:43	REDRES	FIFOPSV
5	6671879.29	P	0	POISSON	02SEP2026	12:43	REDRES	ESTIPSV
6	191258.50	S	0	POISSON	02SEP2026	12:43	REDRES	ESTIPSV

Both come from the same source. *Poulpe* does not use the dissemination weight to expand the sample: it rebuilds the design weights from the auxiliary table that describes the sampling tree, in which the population size of each stage is declared as a count. The rounding makes the reconstructed area weight differ from the one used in R by a factor of 1.0042 here, which propagates to the point estimate. The same factor explains the variance. Applying it to the *Poulpe* total gives 7242263, which is the R estimate.

The gap is below half a percent and has no practical consequence. It is worth knowing only when comparing the two tools digit by digit: the difference lies in how the population sizes are declared to *Poulpe*, not in the variance estimator itself.

Table 4: Variance elements for Poulpe calculated from gustave

Component	typeres	gustave	value
Point estimate	S	est	190449.2
Sampling variance	P	‘var_srs()’ component	6643647.6
F-Non-response component	F		71985478.1
E-Non-response component	E		-71386864.5
Non-response variance	E+F	‘var_pois()’ component	598613.6
?	PF	?	NA
Total variance	FP	var	7242261.2

```
#> Joining with `by = join_by(typeres)`
```

7 Practical Implementation

7.1 Overview

In practice, a variance estimation wrapper must be created for each survey wave that has been extrapolated. One wrapper is needed per level of analysis (individuals, dwellings), since each level involves different stages of the sampling design.

When the sampling design is simple — a (possibly stratified) simple random sample — the `qvar()` function is fully sufficient to create the wrapper. It natively handles stratification, non-response correction and calibration, and its only limitation is that it supports only a single sampling stage.

For multistage designs — the common case for household surveys — `qvar_ms()` extends the same declarative interface to an arbitrary number of stages: stratified simple random sampling at each stage, non-response correction at any subset of stages, calibration at a chosen stage, and configurable treatments for single-unit strata. It also preserves the intermediate information that quality reports and methodological analysis require: variance components by stage, design effects, non-response and domain diagnostics are available at estimation time through the `diagnostics` parameter, and the wrapper itself can be audited with `qvar_notes()` and `inspect_wrapper()`. For most surveys, `qvar_ms()` is therefore the recommended entry point, and this section details all the steps required to build a production-ready wrapper with it, from data preparation through customisation and usage.

Beyond this scope — unequal inclusion probabilities within strata, balanced sampling, shared or integrated weights — `define_variance_wrapper()` can, in principle, handle any situation, and it is the foundation on which both `qvar()` and `qvar_ms()` are built. Writing a wrapper directly with it is, however, demanding in terms of R programming: the calling conventions of the variance function have subtle requirements, and the error messages produced by `gustave` can be difficult to interpret. It should be reserved for designs that `qvar_ms()` cannot describe.

7.2 Data Preparation

Before defining a variance wrapper, the input data must satisfy several conditions. Failing to check these will produce either cryptic errors or, worse, silently incorrect variance estimates. `qvar()` and `qvar_ms()` include a lot of data check to prevent a lot of errors, while `define_variance_wrapper()` needs your own quality checks.

7.2.1 Missing values

Every variable that enters the variance estimation — inclusion probabilities, weights, response indicators, calibration variables — must be checked for missing values. A common symptom of missing values is that the wrapper returns nothing for the point estimate (`est`). This happens when the `reference_weight` vector contains missing values: since the point estimate is computed as a weighted sum using `reference_weight`, a single NA propagates to the output.

```
# Check for NA in critical variables
check_vars <- c("pik_area", "pik_dwel", "w_area", "w_dwel", "prep", "rep")
lfs_samp_dwel %>%
  summarise(across(all_of(check_vars), ~ sum(is.na(.x)))) %>%
  pivot_longer(everything(), names_to = "variable", values_to = "n_missing") %>%
  filter(n_missing > 0)
```

For non-response weights specifically when you use `define_variance_wrapper()`, non-respondents must have a `weight = 0`, not NA. This is because `reference_weight` is set to this weight when defining the wrapper: non-respondents are given weight zero (meaning they are excluded from dissemination), but the value must be present.

```
lfs_samp_dwel <- lfs_samp_dwel %>%
  mutate(w_final_nrc = ifelse(is.na(w_final_nrc), 0, w_final_nrc))
```

7.2.2 Identifier sorting

`gustave` requires that unit identifiers be character vectors, and it issues a warning if they are not sorted. Internally, the variance function receives `y` as a named matrix whose row names correspond to the `reference_id` vector. If the identifiers in the analyst's data are in a different order, `gustave` re-aligns them, but if they are not character vectors, the matching can fail silently.

```
lfs_samp_area <- lfs_samp_area %>%
  mutate(id_area = as.character(id_area)) %>%
  arrange(id_area)

lfs_samp_dwel <- lfs_samp_dwel %>%
  mutate(id_area = as.character(id_area),
         id_dwel = as.character(id_dwel)) %>%
  arrange(id_dwel)

lfs_samp_ind <- lfs_samp_ind %>%
  mutate(id_area = as.character(id_area),
         id_dwel = as.character(id_dwel),
         id_ind = as.character(id_ind)) %>%
  arrange(id_ind)
```

The `arrange()` step is not strictly mandatory but avoids warnings and makes debugging easier when inspecting the `y` matrix inside the variance function.

7.2.3 Confidentiality of technical data

The technical data embedded in the wrapper contains the full sampling information: inclusion probabilities, strata, response probabilities, calibration variables, and unit identifiers. This information is permanently

stored inside the wrapper object. If the wrapper is saved as an `.rds` file and shared with analysts, they gain access to all of this information.

This has implications for data confidentiality. Two strategies can mitigate the risk. First, the technical data can be limited to the strict minimum — only the variables actually needed by the variance function. Second, if the wrapper must be shared externally, consider whether the sampling parameters themselves are sensitive. In most cases, the analyst only needs the wrapper function, not the underlying technical data, but `gustave` does not currently offer a mechanism to encrypt or hide the technical data.

```
# Minimal technical data: only what the variance function needs
technical_data <- list()
technical_data$lfs_samp_area <- lfs_samp_area %>%
  select(id_area, pik_area, w_area)
technical_data$lfs_samp_dwel <- lfs_samp_dwel %>%
  select(id_dwel, id_area, pik, w_area, w_dwel, w_final, prep, rep)
```

7.3 Customising the variance estimation

7.3.1 Customising the Wrapper

The `define_variance_wrapper()` function accepts three types of arguments that are embedded differently in the wrapper.

`technical_data` is a named list of data frames or vectors that the variance function needs. Each element becomes an argument of the variance function with the matching name. This data is fixed at wrapper creation time and cannot be changed by the analyst. Use this for everything that is determined by the survey design: sampling weights, inclusion probabilities, strata, response indicators, and calibration variables. For example, all the names of the parameters and the calibration variables can be set in `technical_data`. This allows the variance function to be reused for another case.

```
var_sample_flex <- function(y, lfs_samp_dwel, cal_vars, id_dwel, stratum,
                           wSmp, prep, wCal) {

  variance <- list()

  resp <- lfs_samp_dwel %>% filter(rep == 1)

  # Use the cal_vars argument instead of extracting variables from lfs_samp_dwel
  cal_form <- as.formula(paste("~", paste(cal_vars, collapse = "+")))
  x_cal <- model.matrix(cal_form, data = resp)

  # Step 1: Replace y with calibration residuals (respondents only)
  y[resp[[id_dwel]], ] <- res_cal(
    y[resp[[id_dwel]], ,drop = FALSE],
    x = x_cal,
    w = resp[[wCal]]
  )

  # Step 2: Non-response variance (Poisson, respondents only)
  variance[["nr"]] <- var_pois(
    y[resp[[id_dwel]], ,drop = FALSE],
    pik = resp[[pik]],
    w = resp[[wSmp]]
  )
}
```

```

# Step 3: Reinroduce non-respondents with zeros
y <- add_zero(y, rownames = lfs_samp_dwel[[id_dwel]])

# Step 4: Sampling variance (SRS, full sample)
variance[["srs"]] <- var_srs(
  y / lfs_samp_dwel[[prep]],
  pik = 1 / lfs_samp_dwel[[wSmp]],
  strata = lfs_samp_dwel[[stratum]]
)

Reduce(`+`, variance)
}

# We need to include the same argument as the variance function

technical_data_flex <- list()
technical_data_flex$lfs_samp_dwel <- lfs_samp_dwel
technical_data_flex$cal_vars <- "id_area"
technical_data_flex$id_dwel <- "id_dwel"
technical_data_flex$wSmp <- "w_final"
technical_data_flex$prep <- "prep"
technical_data_flex$wCal <- "w_final_nrc"
technical_data_flex$stratum <- "id_area"

precision_flex <- define_variance_wrapper(
  variance_function = var_sample_flex,
  technical_data = technical_data_flex,
  reference_id = technical_data_flex$lfs_samp_dwel$id_dwel,
  reference_weight = technical_data_flex$lfs_samp_dwel$w_final_nrc,
  default_id = "id_dwel"
)

precision_flex(lfs_samp_dwel, total(income))
#>
#>      call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 197245302 14044.4 7.374359 162922.6 217975.7

```

technical_param is a named list of parameters that control methodological choices. Like **technical_data**, each element becomes an argument of the variance function. The key difference is that **technical_param** values can be overridden by the analyst at call time, making them ideal for switches such as “with or without calibration” or “alternative non-response model”.

```

var_sample_flex <- function(y, lfs_samp_dwel, cal_vars, id_dwel, stratum,
                           wSmp, prep, wCal, use_calibration = TRUE) {

  variance <- list()

  resp <- lfs_samp_dwel %>% filter(rep == 1)

  if (use_calibration){
    # Use the cal_vars argument instead of extracting variables from lfs_samp_dwel
    cal_form <- as.formula(paste("~", paste(cal_vars, collapse = "+")))
    x_cal <- model.matrix(cal_form, data = resp)
  }
}

```

```

# Step 1: Replace y with calibration residuals (respondents only)
y[resp[[id_dwel]], ] <- res_cal(
  y[resp[[id_dwel]], ,drop = FALSE],
  x = x_cal,
  w = resp[[wCal]]
)
}

# Step 2: Non-response variance (Poisson, respondents only)
variance[["nr"]] <- var_pois(
  y[resp[[id_dwel]], ,drop = FALSE],
  pik = resp[[prep]],
  w = resp[[wSmp]]
)

# Step 3: Reintroduce non-respondents with zeros
y <- add_zero(y, rownames = lfs_samp_dwel[[id_dwel]])

# Step 4: Sampling variance (SRS, full sample)
variance[["srs"]] <- var_srs(
  y / lfs_samp_dwel[[prep]],
  pik = 1 / lfs_samp_dwel[[wSmp]],
  strata = lfs_samp_dwel[[stratum]]
)

Reduce(`+`, variance)
}

technical_data_flex <- list()
technical_data_flex$lfs_samp_dwel <- lfs_samp_dwel
technical_data_flex$cal_vars <- "id_area"
technical_data_flex$id_dwel <- "id_dwel"
technical_data_flex$wSmp <- "w_final"
technical_data_flex$prep <- "prep"
technical_data_flex$wCal <- "w_final_nrc"
technical_data_flex$stratum <- "id_area"

# We need to create another list for technical parameters.
technical_param_flex <- list()
technical_param_flex$use_calibration <- TRUE

precision_flex <- define_variance_wrapper(
  variance_function = var_sample_flex,
  technical_data = technical_data_flex,
  technical_param = technical_param_flex,
  reference_id = technical_data_flex$lfs_samp_dwel$id_dwel,
  reference_weight = technical_data_flex$lfs_samp_dwel$w_final_nrc,
  default_id = "id_dwel"
)

# Analyst can override:
precision_flex(lfs_samp_dwel, total(income)) # with calibration (default)
#>          call n      est variance      std      cv      lower      upper
#> 1 total(y = income) 80 190449.2 197245302 14044.4 7.374359 162922.6 217975.7

```

```
precision_flex(lfs_samp_dwel, total(income), use_calibration = FALSE) # without
  ↪ calibration
#>           call n      est variance      std      cv      lower upper
#> 1 total(y = income) 80 190449.2 524074026 22892.66 12.02035 145580.4 235318
```

objects_to_include is a character vector naming R objects in the current environment that should be included in the wrapper. This is for advanced use cases — for instance, including a custom helper function that the variance function calls internally. In most situations, **technical_data** and **technical_param** are sufficient.

7.3.2 Customising the variance function

The variance function is where the methodological work happens. Its interface must follow a strict pattern: it receives exactly one unnamed “data” argument (*y*, the matrix of variables of interest) and any number of named arguments matching the elements of **technical_data** and **technical_param**.

Input. *y* is a numeric matrix with one column per variable of interest and row names corresponding to the unit identifiers (matching **reference_id**). When the analyst calls **precision(data, total(income))**, the total statistic wrapper extracts the income column, and passes the result as *y*. This means that *y* does not already contain the weighted values.

Output. By default, the function must return the estimated variance for the corresponding variable of interest. If the function returns anything else, **gustave** will raise an error. A customisation of the statistic wrappers allows more options.

Diagnostic output. Within the variance function, **print()** can be used to display intermediate results. These messages appear in the console when the wrapper is called and can be useful for debugging. However, they cannot be captured programmatically by the analyst.

For production use, a better approach is to store diagnostic information in a separate environment that persists across calls:

```
# Create a persistent environment for diagnostics
diag_env <- new.env(parent = emptyenv())

var_sample_diag <- function(y, lfs_samp_area, lfs_samp_dwel, lfs_samp_ind) {
  variance <- list()

  # ... compute variance components ...

  poulpe_F <- 0
  variance[["nr"]] <- 1000
  variance[["srs"]] <- 1000

  # Store diagnostics
  diag_env$components <- variance
  diag_env$poulpe <- tibble(
    typeres = c("FP", "F", "E", "P"),
    value = c(Reduce(`+`, variance),
              poulpe_F,
              variance[["nr"]] - poulpe_F,
              variance[["srs"]])
  )
}
```

```

  Reduce(`+`, variance)
}

technical_data <- list()
technical_data$lfs_samp_area <- lfs_samp_area
technical_data$lfs_samp_dwel <- lfs_samp_dwel
technical_data$lfs_samp_ind <- lfs_samp_ind

precision <- define_variance_wrapper(
  variance_function = var_sample_diag,
  technical_data    = technical_data,
  objects_to_include = "diag_env",
  reference_id      = technical_data$lfs_samp_ind$id_ind,
  reference_weight   = technical_data$lfs_samp_ind$w_final_nrc,
  default_id        = "id_ind"
)

# After calling the wrapper, diagnostics are available:
precision(lfs_samp_ind, total(income))
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 116 190449.2      2000 44.72136 0.02348205 190361.5 190536.8
diag_env$components
#> $nr
#> [1] 1000
#>
#> $srs
#> [1] 1000
diag_env$poulpe
#> # A tibble: 4 x 2
#>   typeres value
#>   <chr>    <dbl>
#> 1 FP      2000
#> 2 F        0
#> 3 E      1000
#> 4 P      1000

```

Note that this approach has a limitation: since `gustave` may call the variance function multiple times (e.g. for domain estimation with `by`), `diag_env` will contain only the results from the last call. For domain estimation, consider accumulating results rather than overwriting.

`qvar_ms()` includes a technical parameter called `diagnostics`. With it, you can obtain Variance components ("`components`"), Design effect ("`deff`"), Degrees of freedom ("`dof`"), Non-response diagnostics ("`nr`") or Domain sample sizes ("`domains`"). See the `qvar_ms()` dedicated vignette for diagnostic explanation.

7.3.3 Customising the statistic wrappers

`gustave` ships with standard statistic wrappers: `total()`, `mean()`, `ratio()`, and `diff_of_means()`. These handle the linearisation step automatically. For instance, `mean()` computes the ratio estimator and its linearised variable, then passes the linearised variable to the variance function.

Additional statistic wrappers can be defined using `define_statistic_wrapper()`. This is useful when the survey requires a non-standard estimator (e.g., a Gini coefficient, a quantile, or a rate with a specific denominator).

```
# Example: a rate estimator (proportion of population in a subgroup)
rate <- define_statistic_wrapper(
  statistic_function = function(y, weight) {
    point <- sum(y * weight) / sum(weight)
    lin <- (y - point) / sum(weight)
    list(point = point, lin = lin)
  },
  arg_type = list(data = "y", weight = "weight")
)
```

The `statistic_function` must return a list with two elements: `point` (the point estimate) and `lin` (the linearised variable). An optional metadata element can carry additional information (e.g. sample size) to the display function.

Custom statistic wrappers are made available inside the variance wrapper through the `objects_to_include` argument:

```
precision <- define_variance_wrapper(
  variance_function = var_sample,
  technical_data = technical_data,
  objects_to_include = "rate",
  reference_id = technical_data$lfs_samp_ind$id_ind,
  reference_weight = technical_data$lfs_samp_ind$w_final_nrc,
  default_id = "id_ind"
)

# Now the analyst can use:
precision(data_mb, rate(unemployed))
```

7.3.4 Customising the display function

By default, `gustave` displays results using `standard_display()`, which shows the estimate, variance, standard error, coefficient of variation, and confidence interval bounds. This default can be replaced when defining a custom statistic wrapper, by passing a `display_function` argument.

A custom display function receives the point estimate, the variance, and any metadata returned by the statistic function. It should return a named vector or a one-row data frame. You can add information that comes from the full estimation pipeline: variance components, Poulpe-equivalent decompositions, sample sizes, or warning flags.

7.4 Multistage variance estimation with `qvar_ms()`

The `qvar()` function performs analytical variance estimation in the most common single-stage case: stratified simple random sampling, non-response correction through reweighting, calibration on margins. `qvar_ms()` extends this logic to **multistage designs**: units of stage 2 are drawn within the units of stage 1, units of stage 3 within the units of stage 2, and so on, with non-response correction possibly occurring at each stage and calibration at one given stage.

The variance is estimated as the sum of the conditional variances of each stage, following the Rao (1975) formula as implemented in the SAS Poulpe macros (Caron, 1998) and natively supported by the variance functions of the `gustave` package: at each stage, the variance contributions (`var_srs()`, `var_pois()`) are *linearly* weighted by the cumulated corrected weight of the stages above, through their `w` argument, and the first-stage variance is computed on the estimated totals per primary sampling unit.

A dedicated vignette *Multistage variance estimation with qvar_ms* walks through **nine cases of increasing complexity**, using the same datasets, to explain how to use `qvar_ms()`.

With `qvar_ms()`, a lot a sampling designs is now covered with a lot of integrity data check and methodological notes. For the most complex or unusual designs (atypical sampling schemes, non-standard non-response models, or experimental weighting procedures) there is little value in building a generic function. In those cases, it is more efficient to work directly with `define_variance_wrapper()` and a custom variance function tailored to the specific situation.

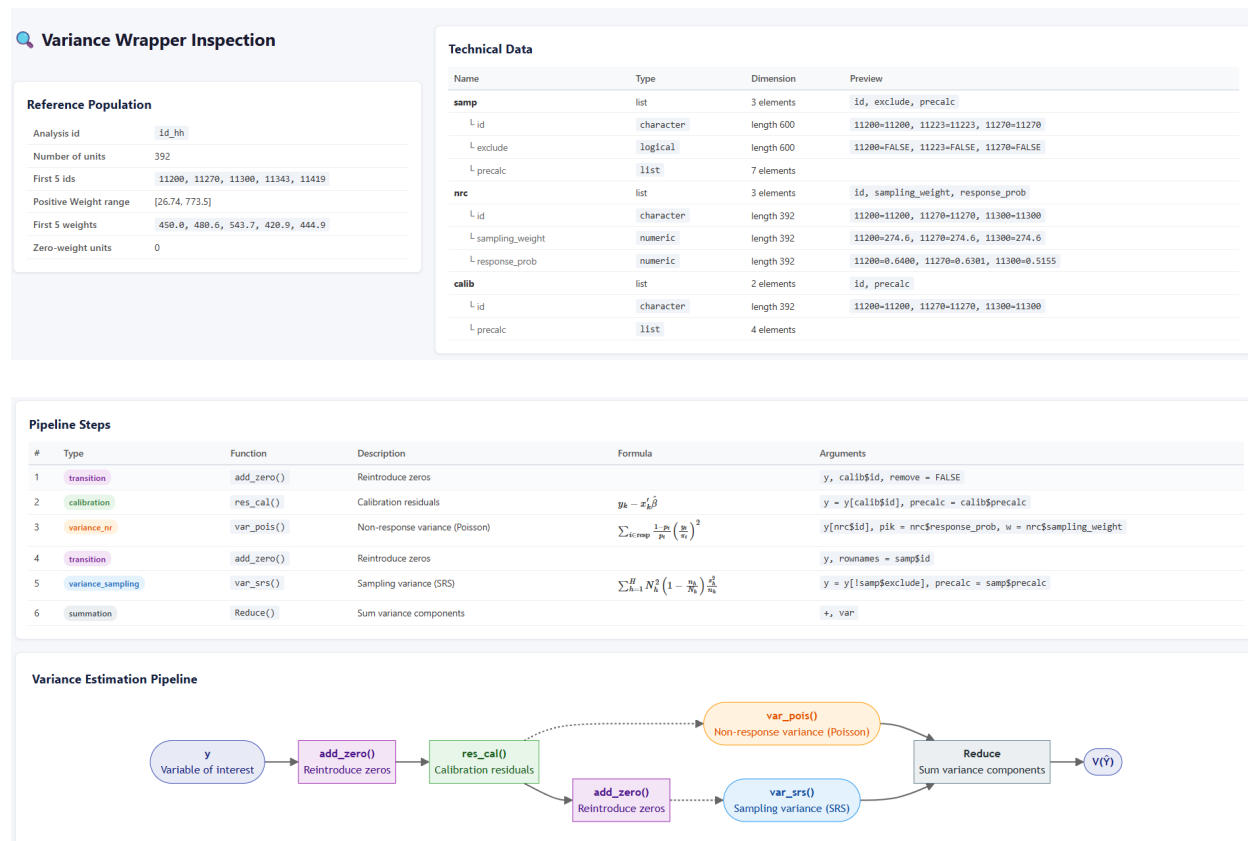
7.5 Inspecting a variance wrapper

Once a variance wrapper has been defined through `qvar()`, `qvar_ms()` or `define_variance_wrapper()`, it can be useful to verify its internal structure before use it in production. The `inspect_wrapper()` function performs this diagnostic by extracting the reference population, the technical data inventory, and the variance estimation pipeline directly from the wrapper object.

This is particularly valuable for documentation and quality reporting: it provides a visual confirmation that the wrapper implements the intended sampling design, with the correct sequence of calibration, non-response correction, aggregation, and variance computation steps. To attach to a quality report or to share with a colleague, `inspect_wrapper()` can generate a self-contained HTML file. Here is an example for the variance function with SRS, non-response and calibration at dwelling level.

`inspect_wrapper()` can also inspect a `qvar_ms()` object. Because of the structure of the `qvar_ms()` object (in list for each stage of sampling), `inspect_wrapper()` call an intern function `inspect_wrapper_ms()` with more information about the wrapper.

```
inspect_wrapper(precision_cal, file = "inspect_precision_cal.html")
```



7.6 Using the Variance Wrapper

Everything in this section applies equally to wrappers created with `qvar()`, `qvar_ms()` and with `define_variance_wrapper()`. Once defined, a wrapper is simply a function that takes a data frame and one or more statistic expressions.

7.6.1 Basic usage

```
# Single statistic
precision_cal(data_hh, total(income))

# Multiple statistics with labels
precision_cal(data_hh,
  "Total income"      = total(income),
  "Mean income"       = mean(income),
  "Unemployment rate" = ratio(unemployed, active)
)
```

7.6.2 Domain estimation with `where` and `by`

The `where` argument restricts estimation to a subpopulation (domain). The `by` argument repeats the estimation for each level of a grouping variable. Both can be combined.

```
# Domain estimation
precision_cal(data_hh, mean(income), where = region == "Flanders")

# Estimation by group
precision_cal(data_hh, mean(income), by = region)

# Combined
precision_cal(data_hh, mean(income), by = sex, where = age_cat == "25-49")
```

7.6.3 Integration with tidyverse workflows

The output of a variance wrapper is a data frame, which makes it straightforward to integrate into tidyverse pipelines. However, because `gustave` wrappers rely on non-standard evaluation (the statistic expression is passed unquoted, e.g. `total(income)` rather than `"total(income)"`), calling them programmatically — in a loop over multiple variables or breakdowns — requires some care.

A practical approach is to build the wrapper call as a string and evaluate it with `eval(parse(text = ...))`. While this is not idiomatic tidyverse style, it is the most reliable way to interact with `gustave`'s non-standard evaluation from within a loop. Two helper functions make this manageable.

`call_gustave()` handles a single call to the wrapper for one indicator and an optional set of grouping variables. When multiple grouping variables are specified (e.g. estimating by region and sex simultaneously), it creates a temporary interaction variable, passes it to the `by` argument, and splits the result back into separate columns afterwards.

```
call_gustave(lfs_samp_dwel, "income", wrapper = "precision_cal", fun_name = "total")
#> Warning: 32 observations do not match any responding units of the survey. They are
↪ discarded.
```

```
#>           call      n      est variance      std      cv      lower      upper
#> 1 total(y = income) 48 190449.2 197245302 14044.4 7.374359 162922.6 217975.7
#>      indic
#> 1 income
```

`batch_gustave()` orchestrates the full estimation process. Given a wrapper, a statistic function name, a list of indicator variables, and optional grouping and domain variables, it runs four successive steps: overall estimates with no breakdown, marginal estimates for each grouping or domain variable separately, crossed estimates combining all grouping variables, and crossed group estimates within each domain. The output is a single tidy data frame with a human-readable `pop` column describing the subpopulation (e.g. “region = Flanders ; sex = M”), ready for export to quality reports.

```
# Example: estimate total and mean for two variables,
# by region and sex (crossed), with id_area as a domain
set.seed(123)
lfs_samp_ind$sex <- sample(c("M", "F"), nrow(lfs_samp_ind), replace=T)

results <- batch_gustave(
  df      = lfs_samp_ind,
  wrapper = "precision_three_2",
  fun_name = "mean",
  vars_indics = c("income", "unemp"),
  vars_groups = "sex",
  vars_domain = "id_area"
)

# The result is a tidy data frame:
results %>%
  group_by(indic) %>%
  slice_min(cv, n = 5) %>%
  select(pop, indic, est, cv, lower, upper)
#> # A tibble: 10 x 6
#> # Groups:   indic [2]
#>   pop      indic      est      cv      lower      upper
#>   <chr>   <chr>    <dbl> <dbl>    <dbl>    <dbl>
#> 1 ALL     income  14.0    1.37  13.6     14.4
#> 2 sex = M income  14.5    2.85  13.7     15.4
#> 3 sex = M income  14.5    2.85  13.7     15.4
#> 4 sex = F income  13.3    3.15  12.5     14.1
#> 5 sex = F income  13.3    3.15  12.5     14.1
#> 6 ALL     unemp    0.103   7.56  0.0876   0.118
#> 7 sex = M unemp    0.108  13.9   0.0786   0.137
#> 8 sex = M unemp    0.108  13.9   0.0786   0.137
#> 9 sex = F unemp    0.0967 15.5   0.0673   0.126
#> 10 sex = F unemp    0.0967 15.5   0.0673   0.126
```

The distinction between `vars_groups` and `vars_domain` is important. Group variables are crossed: if you specify region and sex, you get estimates for every region $\times$ sex combination. Domain variables are not crossed with each other: if you specify `age_cat` as a domain, you get estimates for each age category, and for each age category $\times$ region $\times$ sex combination, but not for age categories crossed with other domain variables. This mirrors the typical structure of quality reports, where a few core breakdowns are fully crossed while additional dimensions are reported separately.

A word of caution when using `batch_gustave()`: since it calls the wrapper repeatedly — once per indicator,

per breakdown, and per step — any `print()` or `message()` statement inside the variance function will be executed at every single call. With a dozen indicators and several grouping variables, this can produce hundreds of lines of console output, making it difficult to spot genuine warnings. It is therefore preferable to remove diagnostic `print()` statements from the variance function before running a batch estimation.

8 Conclusion

This vignette has shown how the **gustave** package can replace the SAS **Poulpe** macro for analytical variance estimation in complex household surveys. Starting from stratified simple random sampling and progressively adding non-response correction, calibration and multistage sampling, we obtained the same results as **Poulpe** — down to a rounding convention on the declared population sizes — within a more flexible and more transparent framework.

The key advantage of **gustave** is its wrapper mechanism. Once the methodologist has encoded the estimation methodology, analysts compute precision measures on their own data without knowing the underlying formulas. The wrapper is a self-contained R object that can be saved, shared and reused across survey waves.

This flexibility has a cost, and it is worth being explicit about it. **Poulpe** imposes a rigid structure: the design is declared as a tree, and the macro either runs or fails. **gustave** asks the methodologist to write the variance decomposition as R code. Every wrapper in this vignette went through several wrong versions before reaching the one printed here. A few examples, all encountered while writing this document: aggregating dwellings to areas with the cumulated weight instead of the conditional one, which inflates the first-stage component by the square of the area weight; reindexing a `sum_by()` output by row position instead of by name, which silently pairs each area total with another area's inclusion probability; weighting the Poisson non-response term quadratically instead of linearly; forgetting to restrict the interest variable to responding units before aggregation, which made the estimated variance depend on whether the analyst had filtered the file beforehand.

None of these mistakes produced an error message. None produced a missing value, a negative variance or an implausible confidence interval. They produced well-formatted numbers, sometimes within a few percent of the correct value, sometimes a hundred times too large. This is the central difficulty of the exercise: a variance estimation error is essentially invisible. A point estimate can be checked against a known total, a previous wave or a published figure. A variance has no such external anchor. It is a number nobody can check by inspection, produced by a formula few people read, and consumed as a confidence interval that will look reasonable whatever its value. We caught our own errors only because we had two independent references — **Poulpe** on the one hand, `qvar()` and `qvar_ms()` on the other hand.

That observation is the reason for `qvar_ms()`. Rather than writing a variance function for each survey, the design is declared argument by argument — one per methodological feature — and the pipeline is derived from the declaration. The estimator is unchanged: the same Rao (1975) decomposition, the same **gustave** variance functions, validated in the dedicated vignette against hand-built wrappers on nine designs of increasing complexity. What changes is the number of opportunities to get it wrong. The weight conventions are checked at definition time rather than assumed; the ordering of the steps is not a matter of writing them in the right sequence; the restriction to responding units is handled internally, so that the result no longer depends on what the analyst passes. The remaining errors are declaration errors — a conditional weight given where a cumulated one is expected — and those are the ones a data check can catch, which is precisely what the calibration g-ratio warning does.

Three further tools support the same goal. Making variance functions configurable through `technical_data` and `technical_param` avoids editing source code from one wave to the next. `qvar_notes()` replays what happened when the wrapper was defined — the coercions, the crossed strata, the excluded single-unit strata. `inspect_wrapper()` rebuilds the pipeline that will actually execute from the technical data of the wrapper itself, and reports it as a document that can be attached to a quality report or handed to a reviewer. Together they make an object that was opaque by construction readable, which is the minimum condition for a second pair of eyes to be of any use.

Taken together, these tools support a clear division of labour: methodologists prepare and document the wrappers, thematic analysts use them as black boxes through `batch_gustave()` to produce quality reports. Wrappers being ordinary R functions, they can also be called from Python through `rpy2`, so that variance estimation fits into existing pipelines without requiring R expertise.

Several topics fall outside this vignette. The designs considered here all rely on simple random sampling, with or without stratification, at each stage. Unequal inclusion probabilities (PPS), balanced sampling (cube method) and systematic sampling call for different estimators — `varDT()` for the Deville-Tillé approximation, `varSYG()` for the Sen-Yates-Grundy estimator — which `gustave` provides and which `define_variance_wrapper()` can accommodate.

We have not compared `gustave` with the `survey` package, which takes the opposite approach: the analyst declares the design once, and the package handles the estimation. Its default treatment of multistage designs is the ultimate cluster approximation, which differs from the stage-by-stage decomposition used here; the size and sign of the difference would be informative in itself. More broadly, the strongest available check on an analytical variance is a replication estimate — jackknife or bootstrap — computed independently. Given how invisible the errors discussed above turned out to be, a second reference is not a refinement but a requirement, and building one is our next step.

Finally, the estimation output has to reach the publication. Packages such as `fonctionr` produce publication-quality graphics with properly computed confidence intervals from `survey` design objects. Bundling a wrapper together with its data into such an object would let precision estimates flow directly into tables and graphs, instead of living in a separate results file.

9 References

- Caron N. (1998), “Le logiciel Poulpe : aspects méthodologiques”, Actes des Journées de méthodologie statistique http://jms-insee.fr/jms1998s03_1/ Deville, J.-C. (1993), Estimation de la variance pour les enquêtes en deux phases, Manuscript, INSEE, Paris.
- Chevalier, Martin, Larbi, Khaled (2018), “gustave: A User-Oriented Statistical Toolkit for Analytical Variance Estimation”, R package
- Deville, J.-C., Tillé, Y. (2005), “Variance approximation under balanced sampling”, Journal of Statistical Planning and Inference, 128, issue 2 569-591
- Rao, J.N.K (1975), “Unbiased variance estimation for multistage designs”, Sankhya, C n°37

10 Annexe

10.1 Poulpe script to variance estimation

The SAS results used as a benchmark throughout this vignette are produced with `%bouclepoulpe_total`, an in-house Statbel macro. `Poulpe` (Caron, 1998) is the reference SAS system for analytical variance estimation developed at Insee. Its power comes at the cost of a verbose interface: a single run requires the sequential invocation of several steps (reading the sampling design tree, computing inclusion probabilities, loading the variables of interest, estimating the variance of totals or of functions of totals), each of which takes several dozen parameters describing the design, the response phases and the `Calmar` calibration. `%bouclepoulpe_total` is nothing more than a convenience layer built on top of this machinery. It performs no variance computation of its own: every estimate it returns is produced by the standard `Poulpe` steps.

10.1.1 Case 2: Adding Non-Response

```
data CALAG.lfs_samp_dwel; set R.lfs_samp_dwel;
NINFFIC = "CA";
respons = rep+1;
run;

DATA BURCE.TREE1;
%ARBREXXX;
infile cards delimiter = "," dsd missover;
input nsup ninf typtir ENTAG ENTI ;
cards;
ZZ,AA,TOT, ,
AA,BA,EXH,id_area, id_area
BA,CA,SAS,id_area,
;
run;

proc sql;
create table BURCE.GEO as select distinct
    id_area, (1) AS Auxniv, sum(w_final) AS NNN
from R.lfs_samp_dwel group by id_area;
quit;

%let mywip = w_dwel_nrc;
%let data_source = CALAG.lfs_samp_dwel;
%let var_out = T1_VARIANCE;
%let myArbinp= TREE1;
%let myDataauxp= GEO;
%let myNbphas= 2;
%let myPipoissp= PREp;
%let myFormulep= POISSON;
%let myEnqredp = ;
%let myExplicp = ;
%let myExplici= ;

%let var1_TST = 1 = 1;
%let var2_TST = 1 = 1;
%let var3_TST = income;

%boclepeoulpe_total(step1=FP,step2=S);
```

10.1.2 Case 3: Adding Calibration

```
data CALAG.lfs_samp_dwel; set R.lfs_samp_dwel;
NINFFIC = "CA";
respons = rep+1;
run;

DATA BURCE.TREE1;
%ARBREXXX;
```

```

infile cards delimiter = "," dsd missover;
input nsup ninf typtir ENTAG ENTI ;
cards;
ZZ,AA,TOT, ,
AA,BA,EXH,id_area, id_area
BA,CA,SAS,id_area,
;
run;

proc sql;
create table BURCE.GEO as select distinct
    id_area, (1) AS Auxniv, sum(w_final) AS NNN
from R.lfs_samp_dwel group by id_area;
quit;

%let mywip = w_dwel_nrc;
%let data_source = CALAG.lfs_samp_dwel;
%let var_out = T2_VARIANCE;
%let myArbinp= TREE1;
%let myDataauxp= GEO;
%let myNbphasp= 2;
%let myPipoissp= PRep;
%let myFormulep= POISSON;
%let myEnqredp = REDRES;
%let myExplicp = id_area;
%let myExpliqp = ;

%let var1_TST = 1 = 1;
%let var2_TST = 1 = 1;
%let var3_TST = income;

%bouclepoulpe_total(step1=FP,step2=S);

```

10.1.3 Case 4: Two-Stage Sampling

```

/***** Paramètres *****/

data CALAG.lfs_samp_dwel; set R.lfs_samp_dwel;
strate_fictive = 1;
respons = rep+1;
NINFFIC = "DA";
run;

proc sql; create table CALAG.lfs_samp_area as select
id_area,mean(1/pik_area) as w_area,
1 as strate_fictive
from R.lfs_samp_area;
quit;

DATA BURCE.TREE2;
%ARBREXXX;
infile cards delimiter = "," dsd missover;

```

```

input nsup ninf typtir ENTAG ENTI ;
cards;
ZZ,AA,TOT, ,
AA,BA,EXH,strate_fictive, strate_fictive
BA,CA,SAS,strate_fictive, id_area
CA,DA,SAS,strate_fictive id_area,
;
run;

proc sql;
create table BURCE.GEO2_A as select distinct
    strate_fictive, (1) AS Auxniv, sum(w_area) AS NNN
from CALAG.lfs_samp_area;

create table BURCE.GEO2_B as select distinct
    strate_fictive, id_area, (2) AS Auxniv, sum(w_dwel) AS NNN
from CALAG.lfs_samp_dwel group by id_area;
quit;

data BURCE.GEO2; set BURCE.GEO2_A BURCE.GEO2_B; run;

%let mywip = w_dwel_nrc;
%let data_source = CALAG.lfs_samp_dwel;
%let var_out = T3_VARIANCE;
%let myArbinp= TREE2;
%let myDataauxp= GEO2;
%let myNbphasp= 2;
%let myPipoissp= PRep;
%let myFormulep= POISSON;
%let myEnqredp = REDRES;
%let myExplicp = id_area;
%let myExliqp = ;

%let var1_TST = 1 = 1;
%let var2_TST = 1 = 1;
%let var3_TST = income;

%bouclepoulpe_total(step1=FP,step2=S);

```

10.1.4 Case 5: Adding individuals answers

```

/***** Paramètres *****/

/***** Paramètres *****/

data CALAG.lfs_samp_ind; set R.lfs_samp_ind;
strate_fictive = 1;
respons = rep+1;
NINFFIC = "EA";
run;

data CALAG.lfs_samp_dwel; set R.lfs_samp_dwel;

```

```

strate_fictive = 1;
run;

proc sql; create table CALAG.lfs_samp_area as select
id_area,mean(1/pik_area) as w_area,
1 as strate_fictive
from R.lfs_samp_area;
quit;

DATA BURCE.TREE3;
%ARBREXXX;
infile cards delimiter = "," dsd missover;
input nsup ninf typtir ENTAG ENTI ;
cards;
ZZ,AA,TOT, ,
AA,BA,EXH,strate_fictive, strate_fictive
BA,CA,SAS,strate_fictive, id_area
CA,DA,SAS,strate_fictive id_area, strate_fictive id_area id_dwel
DA,EA,EXH,strate_fictive id_area id_dwel,
;
run;

proc sql;
create table BURCE.GEO3_A as select distinct
strate_fictive, (1) AS Auxniv, round(sum(w_area)) AS NNN
from CALAG.lfs_samp_area;

create table BURCE.GEO3_B as select distinct
strate_fictive, id_area, (2) AS Auxniv, sum(w_dwel) AS NNN
from CALAG.lfs_samp_dwel group by id_area;

create table BURCE.GEO3_C as select distinct
strate_fictive, id_area, id_dwel, (3) AS Auxniv, sum(w_ind) AS NNN
from CALAG.lfs_samp_ind group by id_dwel;
quit;

data BURCE.GEO3; set BURCE.GEO3_A BURCE.GEO3_B BURCE.GEO3_C; run;

%let mywip = w_dwel_nrc;
%let data_source = CALAG.lfs_samp_ind;
%let var_out = T4_VARIANCE;
%let myArbinp= TREE3;
%let myDataauxp= GEO3;
%let myNbphasp= 2;
%let myPipoissp= PRep;
%let myFormulep= POISSON;
%let myEnqredp = REDRES;
%let myExplicp = id_area;
%let myExpliqp = ;

%let var1_TST = 1 = 1;
%let var2_TST = 1 = 1;
%let var3_TST = income;

```

```
%bouclepoulpe_total(step1=FP,step2=S);
```